Submodularity, Convexity, and Connectivity in Graphs

Lecture Notes on Uncrossing, Orientations, Tree Packings, and Splitting-Off

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Abstract

These notes develop a unified route through fundamental theorems of graph theory—including Hall, Menger, Edmonds, Nash–Williams–Tutte, Robbins, and Nash–Williams' orientation theorem—using one recurrent structural principle: *submodularity*. The graph-theoretic path is organized around *uncrossing*: feasibility is encoded by set-indexed inequalities, equality produces tight sets, and crossing tight sets are replaced by nested, laminar, lattice-like, or partition-like certificates. Eight principal graph-theoretic uncrossing case studies are highlighted in the text: Hall's tight-set induction, the lattice of minimum cuts, Edmonds' arborescence construction, Frank's free-root branching theorem, Frank's rooted-orientation route to the Nash–Williams–Tutte theorem, Mader's degree lemma, Lovász splitting-off, and Mader's balanced-vertex theorem. The first seven are given as complete proofs; the final one is presented at the level of a detailed proof outline. Menger's theorem is proved completely by residual augmentation, and Robbins' theorem is proved completely by a depth-first-search orientation argument. General dual uncrossing and Jain's iterative-rounding application provide two further polyhedral manifestations of the same method.

A complementary path makes the discrete-convex viewpoint explicit through Lovász extensions, the Choquet integral, base polyhedra, polymatroids, normal fans, graphical zonotopes, graph total variation, and submodular flows. The two paths are not parallel stories: the handout marks six explicit *bridge points* where the same object or theorem is visible from both sides. They are (i) cut functions as graph total variation through the coarea formula; (ii) Hall's theorem through transversal matroids and their base polytopes; (iii) minimum cuts through max-flow duality together with cut-polyhedral geometry; (iv) Edmonds' theorem through the arborescence polytope; (v) spanning trees through graphic matroids, tree-packing inequalities, and graphical-zonotope geometry; and (vi) Nash–Williams' orientation theorem through both splitting-off and submodular-flow integrality. To keep the two routes genuinely balanced, the geometric path contains two places where convex geometry supplies a complete optimization argument rather than merely an interpretation: a separation-and-coarea proof of max-flow/min-cut from the directed cut zonotope, and a minimum-norm-base theorem showing how Euclidean projection onto a base polyhedron recovers the minimizers of a submodular function. For graph cuts with vertex prizes, the latter becomes projection onto a translated graphical zonotope. Thus augmentation, threshold integration, supporting geometry, and Euclidean projection become complementary explanations of discrete optimization phenomena.

Splitting-off theorems of Lovász and Mader provide the structural culmination, while the final sections collect classroom problems and research directions, including the edge-flip reconfiguration theory of Ito et al. [35], discrete convex optimization on orientation spaces, and the infinite Nash–Williams orientation problem, for which Assem, Koloschin, and Pitz obtained the optimal bound for locally finite graphs with countably many ends [4].

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Guide to the handout

The material is organized around two complementary ideas that recur throughout modern graph theory and combinatorial optimization.

1. **Uncrossing.** Whenever a connectivity requirement is expressed by a submodular cut function, equality cases tend to be stable under union and intersection. This turns an apparently arbitrary family of obstructions into a laminar, chain-like, or lattice-like family that can be handled by induction and decomposition.

2. **Discrete convexity.** A submodular set function admits a convex, piecewise-linear Lovász extension. On a finite ground set this same level-set formula is the Choquet integral of the set function: the set function plays the role of a possibly nonadditive measure, while a vector $x \in \mathbb{R}^U$ is the real-valued function being integrated. Its base polyhedron is the convex body whose support function is that extension. Many graph-theoretic integrality theorems can therefore be viewed as statements saying that a convex relaxation has integral vertices.

The order of the handout is not meant to be a catalogue of unrelated theorems. It follows a dependency ladder. Part I develops the language—submodularity, tight sets, uncrossing, Lovász extensions, and base polyhedra. Part II begins with Hall's theorem as the simplest complete model of the method, then passes to minimum cuts, directed tree packings, and undirected tree packings. Part III studies orientations and splitting-off, where the same ideas are combined with genuinely local graph transformations. Part IV returns to the more abstract viewpoint: submodular flows package many of the preceding cut systems into a common polyhedral framework, and several further examples show how far the same mathematics extends.

This organization separates two useful reading paths. A reader primarily interested in graph-theoretic proofs may read the elementary submodularity and uncrossing material first and then move directly to Hall, Menger, branchings, tree packings, orientations, and splitting-off. A reader interested in the convex-geometric viewpoint should also follow the Lovász extension, base-polyhedron, zonotope, and submodular-flow sections. The two paths meet repeatedly; cross-references indicate those meeting points.

A second hierarchy also runs through the notes: matroid rank functions are special integral polymatroid rank functions, polymatroid ranks are special monotone submodular functions, and graph cut functions belong to the still larger class of general submodular functions. Short signposts later in the handout indicate when an argument is genuinely matroidal, when it belongs to the polymatroid level, and when monotonicity has already disappeared and only general submodularity remains.

Three kinds of highlighted signposts. To make the architecture visible while reading, the handout uses three recurring boxes.

- An orange **UNCROSSING CASE STUDY** box marks one of the eight principal graph-theoretic uses of the method. The box also says whether the text gives a complete proof or a proof strategy/scheme.
- A violet **BRIDGE** box marks one of six places where the graph-theoretic and convex-geometric reading paths explicitly meet. The six bridge topics are: cuts/coarea, Hall/transversal matroids, minimum cuts/cut polyhedra, arborescences/arborescence polytopes, spanning trees/graphic-matroid–zonotope geometry, and Nash–Williams orientations/splitting-off versus submodular-flow integrality.
- A cyan **GEOMETRIC PROOF** box marks a place where the convex-geometric language is not merely interpretive but supplies a complete proof. Two principal instances are developed: separation plus coarea for max-flow/min-cut, and Euclidean projection onto a base polyhedron for submodular minimization.

The counts refer to the main narrative, not to every local use of submodularity. For example, the laminar-dual principle and iterative rounding also use uncrossing, but they are treated as general polyhedral mechanisms rather than additional graph-theorem case studies.

In particular, Hall's theorem should be viewed as a *prototype*, not as the first item in a flat list of applications. The later applications are therefore not placed one after another immediately after Hall merely because they share its proof pattern. They are ordered so that each new section introduces the extra structure needed by the next one: residual cuts, arborescences, partition inequalities, orientations, splitting-off, and finally the general submodular-flow formalism.

Throughout the handout, graph and digraph theorems are stated for finite loopless graphs or multigraphs; parallel edges and arcs are allowed and are counted with multiplicity. For a digraph, "k-arc-strong" means that every nonempty proper vertex set has at least k entering arcs (equivalently, at least k leaving arcs). For an undirected graph, "k-edge-connected" means that every nontrivial cut has size at least k. We use $\rho_D(X)$ for the number of arcs entering X and $\delta_D^+(X)$ for the number of arcs leaving X; thus an orientation is k-arc-strong exactly

when $\rho_D(X), \delta_D^+(X) \geq k$ for every $\varnothing \neq X \subsetneq V$. In Frank's theorem in Section 6, the spanning arborescences may have different roots. The inverse of complete splitting-off in a $2k$-edge-connected graph is described as *pinching k edges*. These conventions will be used consistently below.

Part I
Foundations: submodularity, uncrossing, and convex geometry

1 Why submodularity is a graph-theoretic form of convexity

1.1 Set functions and diminishing returns

Let U be a finite ground set. A function $f : 2^U \to \mathbb{R}$ is *submodular* if $f(X) + f(Y) \geq f(X \cap Y) + f(X \cup Y)$ for all $X, Y \subseteq U$. It is *modular* if equality always holds, and *supermodular* if the inequality is reversed.

The most useful local form of the definition is the following diminishing-returns condition. Importantly, no normalization assumption is needed.

Proposition (Equivalent forms of submodularity)**.** *For an arbitrary set function $f : 2^U \to \mathbb{R}$, the following are equivalent.*

(i) f is submodular, meaning that $f(X) + f(Y) \geq f(X \cap Y) + f(X \cup Y)$ for all $X, Y \subseteq U$.

(ii) f has diminishing marginal returns*:*

$$\Delta_e f(A) := f(A \cup \{e\}) - f(A) \geq f(B \cup \{e\}) - f(B) =: \Delta_e f(B)$$

whenever $A \subseteq B$ and $e \notin B$.

(iii) All discrete mixed second differences are nonpositive: for distinct $e, g \notin A$,

$$\Delta_g \Delta_e f(A) := f(A \cup \{e, g\}) - f(A \cup \{g\}) - f(A \cup \{e\}) + f(A) \leq 0.$$

Proof. Assume (i). Apply submodularity to $A \cup \{e\}$ and B. Since $(A \cup \{e\}) \cap B = A$ and $(A \cup \{e\}) \cup B = B \cup \{e\}$, we get $f(A \cup \{e\}) - f(A) \geq f(B \cup \{e\}) - f(B)$, which is (ii). Taking $B = A \cup \{g\}$ shows that (ii) implies (iii).

Conversely, assume (iii). If $B \setminus A = \{g_1, \ldots, g_m\}$, set $A_j = A \cup \{g_1, \ldots, g_j\}$. Repeatedly applying (iii) gives

$$\Delta_e f(A) = \Delta_e f(A_0) \geq \Delta_e f(A_1) \geq \cdots \geq \Delta_e f(A_m) = \Delta_e f(B),$$

so (ii) holds. Finally, to derive (i) from (ii), write $X \setminus Y = \{e_1, \ldots, e_k\}$. For each i, put $A_i := (X \cap Y) \cup \{e_1, \ldots, e_{i-1}\}$ and $B_i := Y \cup \{e_1, \ldots, e_{i-1}\}$. Then $A_i \subseteq B_i$ and $e_i \notin B_i$, so summing $\Delta_{e_i} f(A_i) \geq \Delta_{e_i} f(B_i)$ telescopes to $f(X) - f(X \cap Y) \geq f(X \cup Y) - f(Y)$, which is exactly the submodular inequality. □

Remark (Normalization is a separate convention). The condition $f(\varnothing) = 0$ is *not* part of the equivalence above. Indeed, $f^\circ(X) := f(X) - f(\varnothing)$ is normalized and has exactly the same marginal differences and the same submodularity inequalities as f. Thus one may normalize a function whenever convenient without changing its submodular structure. Normalization becomes useful later when we write the Lovász extension and the base polyhedron in their cleanest standard form.

Proposition (A small calculus of submodular functions)**.** *Let $f, g : 2^U \to \mathbb{R}$ be submodular.*

(a) If $a, b \geq 0$, then $af + bg$ is submodular.

(b) If m is modular, then $f + m$ is submodular and has exactly the same submodularity slack as f.

(c) For $W \subseteq U$, the restriction $X \mapsto f(X)$ on 2^W is submodular; for fixed $A \subseteq U$, the contraction $f/A(X) := f(X \cup A) - f(A)$, defined for $X \subseteq U \setminus A$, is submodular.

(d) The complementary function $f^*(X) := f(U \setminus X)$ is submodular.

(e) A set function m is modular if and only if there are constants c and w_u such that $m(X) = c + \sum_{u \in X} w_u$.

Proof. Parts (a)–(d) follow by substituting into the four-set inequality. For (e), one direction is immediate. Conversely, if m is modular, put $c = m(\varnothing)$ and $w_u = m(\{u\}) - m(\varnothing)$. Modularity applied repeatedly to a set and a singleton shows by induction on $|X|$ that $m(X) = c + \sum_{u \in X} w_u$. □

These elementary facts are used constantly, often without comment. Standard references for the basic theory include Fujishige [27], Lovász [41], and Schrijver [60].

Example 1.1 (Cardinality)**.** The function $X \mapsto |X|$ is modular. More generally, every weighted cardinality function $m(X) = \sum_{u \in X} w_u$ is modular.

Example (Concave functions of cardinality)**.** Let $\phi : \{0, 1, \ldots, |U|\} \to \mathbb{R}$ have decreasing successive differences, $\phi(j+1) - \phi(j) \geq \phi(j+2) - \phi(j+1)$. Then $f(X) = \phi(|X|)$ is submodular. Indeed, $\Delta_e f(X) = \phi(|X| + 1) - \phi(|X|)$ decreases as $|X|$ grows. Thus ordinary concavity on the integer line is already a source of submodularity. The uniform-matroid rank function $X \mapsto \min\{|X|, r\}$ is the basic example.

Example 1.2 (Neighborhoods)**.** Let $G = (L, R; E)$ be bipartite and let $N(X) \subseteq R$ denote the neighborhood of $X \subseteq L$. Then $\gamma(X) = |N(X)|$ is submodular because $N(X \cup Y) = N(X) \cup N(Y)$ while $N(X \cap Y) \subseteq N(X) \cap N(Y)$. More precisely, the entire slack is visible:

$$\gamma(X) + \gamma(Y) - \gamma(X \cap Y) - \gamma(X \cup Y) = |N(X) \cap N(Y)| - |N(X \cap Y)| \geq 0.$$

Thus equality means that every right vertex adjacent to both X and Y is already adjacent to a vertex in $X \cap Y$.

Example 1.3 (Undirected cuts)**.** For an undirected graph $G = (V, E)$, let $d_G(X) = |\delta_G(X)|$, where $\delta_G(X)$ consists of edges with exactly one endpoint in X. Then d_G is symmetric, $d_G(X) = d_G(V \setminus X)$, and submodular. In fact, if $P = X \setminus Y$ and $Q = Y \setminus X$, then the exact identity is

$$d_G(X) + d_G(Y) - d_G(X \cap Y) - d_G(X \cup Y) = 2|E(P, Q)|. \tag{1}$$

Likewise, writing $C = X \cap Y$ and $D = V \setminus (X \cup Y)$,

$$d_G(X) + d_G(Y) - d_G(X \setminus Y) - d_G(Y \setminus X) = 2|E(C, D)|. \tag{2}$$

The second identity is the *posimodular* inequality. Both formulas remain true for nonnegative edge weights after replacing cardinalities by total weights.

Proposition (Symmetry turns submodularity into posimodularity)**.** *If $f : 2^U \to \mathbb{R}$ is symmetric, meaning $f(X) = f(U \setminus X)$, and submodular, then $f(X) + f(Y) \geq f(X \setminus Y) + f(Y \setminus X)$ for all $X, Y \subseteq U$.*

Proof. Apply submodularity to X and $U \setminus Y$. Their intersection is $X \setminus Y$, while the complement of their union is $Y \setminus X$; now use symmetry. □

Example 1.4 (Directed in-cuts)**.** For a digraph $D = (V, A)$ define $\rho_D(X) := |\{uv \in A : u \notin X,\ v \in X\}|$. Then ρ_D is submodular. Again there is an exact slack formula. If $P = X \setminus Y$ and $Q = Y \setminus X$, then

$$\rho_D(X) + \rho_D(Y) - \rho_D(X \cap Y) - \rho_D(X \cup Y) = |A(P, Q)| + |A(Q, P)|, \tag{3}$$

where $A(P, Q)$ denotes arcs directed from P to Q. Thus the failure of equality is measured exactly by arcs running between the two "crossing pieces". This directed cut function will drive Menger's and Edmonds' theorems. Notice that ρ_D is generally not monotone, so although it is submodular it is not, in general, a polymatroid rank function.

Example 1.5 (Matroids: independent sets and rank)**.** A matroid is a finite set system that abstracts the familiar notion of linear independence. It is useful here because it can be described in two equivalent ways, one combinatorial and one numerical.

Independent-set description. Let U be a finite ground set. A family $\mathcal{I} \subseteq 2^U$ is the family of *independent sets* of a matroid if

(i) $\varnothing \in \mathcal{I}$;

(ii) if $J \subseteq I \in \mathcal{I}$, then $J \in \mathcal{I}$;

(iii) if $I, J \in \mathcal{I}$ and $|I| < |J|$, then some $e \in J \setminus I$ satisfies $I \cup \{e\} \in \mathcal{I}$.

A *basis* of a set $X \subseteq U$ is a maximal independent subset of X. The exchange axiom implies that all bases of X have the same size.

Rank-function description. Equivalently, a matroid may be specified by an integer-valued function $r : 2^U \to \mathbb{Z}_{\geq 0}$ satisfying

(i) $0 \leq r(X) \leq |X|$ for every $X \subseteq U$;

(ii) $X \subseteq Y$ implies $r(X) \leq r(Y)$;

(iii) $r(X) + r(Y) \geq r(X \cap Y) + r(X \cup Y)$ for all $X, Y \subseteq U$.

Thus a matroid rank function is, in particular, a normalized, monotone, integer-valued submodular function.

Here is the equivalence in brief. Starting from the independent sets, define

$$r_M(X) := \max\{|I| : I \subseteq X,\ I \in \mathcal{I}\}.$$

Then $r_M(X)$ is the size of any basis of X. The first two rank axioms are immediate. To see submodularity, choose a basis I of $X \cap Y$, extend it to a basis $I \cup J$ of X, and then extend this to a basis $I \cup J \cup K$ of $X \cup Y$, where $K \subseteq Y \setminus X$. Since $I \cup K$ is independent in Y,

$$r_M(Y) \geq |I| + |K|,$$

and hence

$$r_M(X) + r_M(Y) \geq (|I| + |J|) + (|I| + |K|) = r_M(X \cap Y) + r_M(X \cup Y).$$

Conversely, suppose r satisfies the three rank axioms and define

$$\mathcal{I}_r := \{I \subseteq U : r(I) = |I|\}.$$

These are the independent sets. Indeed, submodularity implies that adding one element can increase rank by at most one. Consequently every subset of a set in $\mathcal{I}_r$ is again in $\mathcal{I}_r$. If $I, J \in \mathcal{I}_r$

with $|I| < |J|$ and no element of $J \setminus I$ could be added to I while preserving independence, then every such element would have zero rank gain at I. By diminishing returns, it would also have zero rank gain after any further additions, giving $r(I \cup J) = r(I) = |I|$, contrary to $r(I \cup J) \geq r(J) = |J|$. Thus the exchange axiom follows.

A concrete example is the *graphic matroid*: U is the edge set of a graph, and a set of edges is independent exactly when it contains no cycle. Its bases are maximal forests (spanning trees when the graph is connected). In this example, the exchange property for forests and the submodularity of the corresponding rank function are two forms of the same structure.

The point for the present notes is therefore that submodularity is the numerical shadow of matroid exchange.

From matroids to polymatroids to general submodularity. The rank axioms above sit inside a useful hierarchy that will reappear throughout the handout. A *polymatroid rank function* is a normalized, monotone, submodular function $p : 2^U \to \mathbb{R}_{\geq 0}$; it is an *integral polymatroid rank function* if it is integer-valued. Matroid rank functions are precisely those integral polymatroid rank functions for which $p(\{e\}) \leq 1$ for every element e. Indeed, submodularity then gives diminishing returns, so

$$p(X) \leq \sum_{e \in X} p(\{e\}) \leq |X|,$$

and the matroid rank axioms are recovered.

Thus one should keep in mind the hierarchy

$$\boxed{\begin{array}{c}\text{matroid ranks} \subset \text{integral polymatroid ranks} \\ \subset \text{polymatroid ranks} \subset \text{normalized submodular functions.}\end{array}}$$

The last inclusion is strict because a general submodular function need not be monotone. This distinction is important in graph theory: the neighborhood function $X \mapsto |N(X)|$ in Hall's theorem is an integral polymatroid rank function, whereas graph cut functions and directed in-cut functions are submodular but generally not monotone, so they live in the larger submodular world.

For a tiny nonmatroidal polymatroid example, choose integers $c_e \geq 0$ and set

$$p(X) = \min\left\{5, \sum_{e \in X} c_e\right\}.$$

This is normalized, monotone, integer-valued, and submodular. If some $c_e > 1$, then one element can contribute more than one unit of rank, so p is an integral polymatroid rank function but not a matroid rank function. Later sections will repeatedly indicate which level of this hierarchy is actually being used.

1.2 Tight sets and the basic uncrossing lemma

Equality cases of submodular inequalities will play a central role. We use the standard term *tight sets* for sets on which the relevant lower bound is attained.

Definition 1.6 (Tight set)**.** Let f be submodular and m modular, and suppose $f(X) \geq m(X)$ for all X in the domain of interest. A set X is *tight* if $f(X) = m(X)$.

For later use it is helpful to name the amount by which submodularity is strict: set $\sigma_f(X, Y) := f(X) + f(Y) - f(X \cap Y) - f(X \cup Y)$, so submodularity says $\sigma_f(X, Y) \geq 0$. For

graph cuts, (1) and (3) show that σ_f is not an abstract defect: it literally counts edges or arcs between the two crossing pieces.

Lemma 1.7 (Uncrossing of tight sets)**.** *If X and Y are tight and the four sets involved remain in the domain, then $X \cap Y$ and $X \cup Y$ are tight. Moreover, the submodularity slack vanishes: $\sigma_f(X, Y) = 0$.*

Proof. We have

$$m(X) + m(Y) = f(X) + f(Y) \geq f(X \cap Y) + f(X \cup Y) \geq m(X \cap Y) + m(X \cup Y).$$

The first and last expressions are equal by modularity of m, hence equality holds throughout. This proves both tightness of the intersection and union and vanishing of the submodularity slack. □

Corollary (Lattice of minimizers)**.** *If $f : 2^U \to \mathbb{R}$ is submodular and $\mu = \min_{X \subseteq U} f(X)$, then the minimizer family $\arg\min f := \{X : f(X) = \mu\}$ is closed under union and intersection. In particular, it has a unique inclusion-minimal member and a unique inclusion-maximal member.*

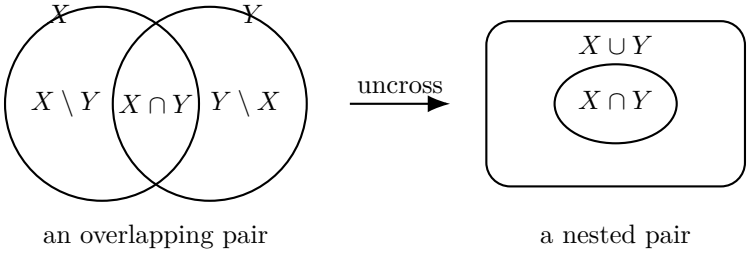
Proof. Apply Lemma 1.7 with the constant modular function $m(X) = \mu$. Finite closure under intersection and union gives the two extremal minimizers. □

Corollary (Zero-slack structure for tight graph cuts)**.** *Suppose a cut function is compared with a modular lower bound and X, Y are tight. For an undirected graph, the total weight of edges between $X \setminus Y$ and $Y \setminus X$ is zero. For a digraph, the total weight of arcs in either direction between these two pieces is zero. With strictly positive edge or arc weights, there are no such edges or arcs at all.*

Proof. Combine Lemma 1.7 with the exact slack identities (1) and (3). □

Convexity lens. For a convex function, equality in Jensen's inequality is highly structured: it often signals that the function is affine on the segment joining two points. In submodular combinatorics, tight sets play an analogous role. When two tight sets overlap nonlaminarily, the submodular inequality forces new equality cases at their union and intersection. Repeatedly replacing an overlapping pair by its intersection and union is the discrete analogue of exposing a face of a convex body.

A family $\mathcal{L} \subseteq 2^U$ is *laminar* if every two members are nested or disjoint: for $X, Y \in \mathcal{L}$, either $X \subseteq Y$, or $Y \subseteq X$, or $X \cap Y = \varnothing$. Thus two sets violate laminarity precisely when $X \cap Y \neq \varnothing$ and neither contains the other. We shall call such a pair an *overlapping pair*. Some authors use the word "crossing" for this configuration, while others additionally require $U \setminus (X \cup Y) \neq \varnothing$; spelling out the convention avoids a common source of confusion.



an overlapping pair a nested pair

Figure 1: The elementary uncrossing move replaces an overlapping pair by its intersection and union. When X and Y are tight, submodularity keeps both replacement sets tight.

For $X \subseteq U$, let $\mathbf{1}_X \in \{0,1\}^U$ denote its incidence vector. If $X_1, \dots, X_r$ is a list of sets, then the u-coordinate of $\mathbf{1}_{X_1} + \cdots + \mathbf{1}_{X_r}$ is simply the number of sets X_i that contain u. Thus equality of two such sums means: *every element of U occurs with the same total multiplicity in the two collections.* This is exactly the piece of data that the next proposition preserves.

Proposition 1.8 (Laminarization principle—in fact, chainification)**.** *Assume that the hypotheses of Lemma 1.7 hold for all subsets of the finite ground set U. Let $\mathcal{F} = (X_1, \dots, X_r)$ be a finite multiset of tight sets, and for $u \in U$ put $c(u) := |\{i : u \in X_i\}|$. For $k = 1, \dots, r$ define the threshold set*

$$L_k := \{u \in U : c(u) \geq k\}. \tag{4}$$

Then each L_k is tight and

$$L_1 \supseteq L_2 \supseteq \cdots \supseteq L_r.$$

Moreover,

$$\sum_{k=1}^{r} \mathbf{1}_{L_k} = \sum_{i=1}^{r} \mathbf{1}_{X_i}. \tag{5}$$

Hence the original tight sets may be replaced not merely by a laminar multiset, but by a nested chain of tight sets, without changing how many times each element of U is counted.

Proof. The sets L_k are nested directly from their definition: if an element occurs in at least $k+1$ of the X_i, then it certainly occurs in at least k of them.

Next fix $u \in U$. The u-coordinate of the right-hand side of (5) is $c(u)$. On the left-hand side, u belongs to exactly the sets $L_1, L_2, \dots, L_{c(u)}$, and therefore is again counted exactly $c(u)$ times. This proves (5).

It remains only to check that every L_k is tight. An element belongs to at least k of the sets $X_1, \dots, X_r$ if and only if it belongs to all the sets indexed by at least one k-element subset of $\{1, \dots, r\}$. Hence

$$L_k = \bigcup_{\substack{I \subseteq \{1,\dots,r\} \\ |I|=k}} \bigcap_{i \in I} X_i. \tag{6}$$

By Lemma 1.7, the family of tight sets is closed under pairwise intersection and union. By induction it is therefore closed under all finite intersections and finite unions. Formula (6) now shows that L_k is tight. □

The case of two sets: the basic uncrossing move. For $r = 2$, the threshold sets are $L_1 = X_1 \cup X_2$ and $L_2 = X_1 \cap X_2$. Thus Proposition 1.8 reduces to the familiar replacement of (X, Y) by $(X \cap Y, X \cup Y)$. For example, if $X = \{1, 2\}$ and $Y = \{2, 3\}$, then the two original sets count the elements $1, 2, 3$ with multiplicities $(1, 2, 1)$, and the uncrossed pair $\{2\}, \{1, 2, 3\}$ has exactly the same multiplicities. In vector form, $\mathbf{1}_{\{1,2\}} + \mathbf{1}_{\{2,3\}} = \mathbf{1}_{\{2\}} + \mathbf{1}_{\{1,2,3\}}$. This elementary identity is the local mechanism behind the proposition.

Remark 1.9 (Why this is called uncrossing). The preceding proof gives a canonical answer immediately: sort the elements by their multiplicities and take the level sets L_k. The traditional "uncrossing proof" reaches a laminar family by repeatedly performing the local move $(X, Y) \longmapsto (X \cap Y,\ X \cup Y)$ on an overlapping pair. Lemma 1.7 preserves tightness, and coordinatewise one has $\mathbf{1}_X + \mathbf{1}_Y = \mathbf{1}_{X \cap Y} + \mathbf{1}_{X \cup Y}$. To see that this process cannot continue forever, among all representations with the same aggregate incidence vector choose one maximizing $\Phi = \sum_Z |Z|^2$. If X, Y overlap, uncrossing changes the potential by

$$|X \cap Y|^2 + |X \cup Y|^2 - |X|^2 - |Y|^2 = 2|X \setminus Y|\,|Y \setminus X| > 0,$$

a contradiction. Notice the direction: the usual quadratic potential must be *maximized*, not minimized.

Remark 1.10 (What the proposition does *not* say). It does not say that the family of *all* tight sets is laminar. In fact, Lemma 1.7 says that the tight sets form a sublattice of 2^U: they are closed under union and intersection, and such a lattice may contain many overlapping pairs. The conclusion is instead that, whenever a proof only needs a finite collection of tight sets through the aggregate vector $\sum_i \mathbf{1}_{X_i}$, that collection can be replaced by a particularly simple one—indeed, by the chain in Proposition 1.8.

Relation with the more familiar LP uncrossing principle. There is a closely related but genuinely different statement that appears in cut-based linear programming. There the relevant vector attached to a set $S \subseteq V$ is often the *edge-cut incidence vector* $\mathbf{1}_{\delta(S)}$, not the vertex incidence vector $\mathbf{1}_S$. Moreover, the right-hand side of the cut constraints may be only intersecting-supermodular or skew-supermodular. In that setting, uncrossing usually proves that one can choose a *laminar* (or cross-free) family of linearly independent tight constraints; one should not expect the stronger chain conclusion above. In survivable network design, for example, two crossing tight cuts may have to be replaced either by $A \cap B, A \cup B$ or by $A \setminus B, B \setminus A$, depending on which supermodular inequality is available. This is the form of uncrossing used in iterative rounding and related network-design arguments. The weighted LP version will be developed later in Proposition 2.18.

Further reading. For the foundations of submodular polyhedra, see Edmonds' seminal paper [13] and Lovász's convex viewpoint [41]. For a systematic treatment of submodular systems, base polyhedra, and intersecting/crossing submodularity, see Fujishige [27, Chapter 2]; Schrijver's Chapter 44 is another standard entry point [60, Chapter 44]. For the graph-theoretic side, Frank's survey [20] and monograph [22] explain why uncrossing recurs throughout connectivity theory. For readers who want to see the laminar-tight-constraint principle used concretely inside LP arguments, Lau–Ravi–Singh introduce uncrossing already in their spanning-tree chapter and use it repeatedly later [39, Chapter 4]; Jain's survivable-network theorem is a classical application [36]. A modern chapter devoted specifically to "duality, cuts, and uncrossing" is given by Traub and Vygen [61, Chapter 4].

1.3 The Lovász extension

This is the first point at which normalization is genuinely convenient. As noted above, no generality is lost: replace f by $f - f(\varnothing)$ if necessary.

Before introducing the extension, let us record the continuous notion that will be used. No background in convex analysis is assumed.

Definition (Convex function)**.** A function $\Phi : \mathbb{R}^n \to \mathbb{R}$ is *convex* if for every $x, z \in \mathbb{R}^n$ and every $\lambda \in [0, 1]$,

$$\Phi(\lambda x + (1-\lambda)z) \leq \lambda\Phi(x) + (1-\lambda)\Phi(z). \tag{7}$$

Thus the value of Φ at a point on the line segment joining x and z is no larger than the corresponding linear interpolation of the endpoint values. A linear (indeed, affine) function satisfies (7) with equality.

The geometric object attached to a function is its *epigraph*

$$\operatorname{epi}(\Phi) := \{(x, t) \in \mathbb{R}^n \times \mathbb{R} : t \geq \Phi(x)\}.$$

The defining inequality for convexity is exactly the statement that this set is convex.

Proposition 1.11 (Convexity and epigraphs)**.** *A function $\Phi : \mathbb{R}^n \to \mathbb{R}$ is convex if and only if its epigraph* $\operatorname{epi}(\Phi)$ *is a convex subset of* $\mathbb{R}^{n+1}$.

Proof. Assume first that Φ is convex. If (x, a) and (z, b) lie in the epigraph, then $a \geq \Phi(x)$ and $b \geq \Phi(z)$; hence

$$\lambda a + (1-\lambda)b \geq \lambda\Phi(x) + (1-\lambda)\Phi(z) \geq \Phi(\lambda x + (1-\lambda)z),$$

so the whole segment stays in the epigraph. Conversely, apply convexity of the epigraph to the two boundary points $(x, \Phi(x))$ and $(z, \Phi(z))$; the resulting point must lie above the graph of Φ, which is exactly (7). □

This immediately explains another fact used repeatedly below. If $\Phi(x) = \sup_{a\in A} \ell_a(x)$ for affine functions ℓ_a, then

$$\operatorname{epi}(\Phi) = \bigcap_{a\in A} \{(x,t) : t \geq \ell_a(x)\},$$

an intersection of halfspaces. Hence Φ is convex. In particular, the pointwise maximum of finitely many linear functions is convex. We shall use this geometric observation rather than reprove the same inequality each time a support function appears.

For a nonempty set $K \subseteq \mathbb{R}^n$, its *support function* is

$$h_K(c) := \sup_{y\in K} c^T y.$$

It is therefore convex as a function of c. When K is compact, the supremum is a maximum.

The second elementary geometric fact we need is separation.

Theorem 1.12 (Separating a point from a closed convex set)**.** *Let $K \subseteq \mathbb{R}^n$ be nonempty, closed, and convex, and let $p \notin K$. Then there exist $c \neq 0$ and $\alpha \in \mathbb{R}$ such that*

$$c^T p > \alpha \geq c^T z \qquad (z \in K).$$

If K is compact, this may be written as

$$c^T p > h_K(c) = \max_{z\in K} c^T z.$$

Proof. Let $q \in K$ be a point nearest to p and put $c = p - q$. For every $z \in K$, the segment $q + t(z - q)$ lies in K for $0 \leq t \leq 1$. Since $t = 0$ minimizes the squared distance from p along this segment, the right derivative at 0 is nonnegative, which gives $c^T(z - q) \leq 0$. Thus $c^T z \leq c^T q < c^T p$, because $c \neq 0$. Take $\alpha = c^T q$. This nearest-point proof will reappear later in the minimum-norm-base theorem. □

For standard convex-analysis background, including epigraphs, supporting hyperplanes, and separation, see Rockafellar [55]. The important point for us is that submodularity will make these continuous separating and supporting objects collapse back to remarkably concrete subset certificates.

From integrating indicator functions to integrating a set function. There is a useful measure-theoretic way to anticipate the definition. If μ is an ordinary finite measure on U, then the set function is recovered from integration of indicators:

$$\mu(A) = \int_U \mathbf{1}_A \, d\mu.$$

The ordinary integral extends these values from 0–1-valued functions to arbitrary real-valued functions. On a finite ground set, a vector $x \in \mathbb{R}^U$ is simply a real-valued function $u \mapsto x_u$. The Lovász extension performs an analogous extension for a general set function f: it starts from $f(A) = \widehat{f}(\mathbf{1}_A)$ and assigns a value to every real vector x.

When f is monotone and normalized, it is called a *capacity* in the terminology of nonadditive measure theory. The resulting level-set integral is the finite *Choquet integral*.

Thus one may read the same formula in two languages:

Lovász extension in submodular optimization	=	Choquet integral on a finite space.

The additive case is the familiar one: if $f(A) = \sum_{i \in A} p_i$, then the extension is just the ordinary integral $\sum_i p_i x_i$. Submodularity measures how far one may move away from additivity while retaining convexity of this extension.

Remark (A parallel passage: from finite graphs to graphons). There is a useful parallel, although it is important not to confuse it with the Lovász extension itself. A finite simple graph G on n vertices can be represented by its adjacency matrix, hence by a symmetric 0–1 step function $W_G : [0,1]^2 \to \{0,1\}$ obtained by dividing $[0,1]$ into n equal intervals and making W_G constant on each rectangle. In the limit theory of dense graphs one allows instead a symmetric measurable kernel

$$W : [0,1]^2 \longrightarrow [0,1],$$

called a *graphon*. Convergent dense graph sequences admit such measurable limit objects, and subgraph densities become genuine multiple integrals of products of W; see Lovász–Szegedy [42] and Lovász [43].

The analogy with the present discussion is conceptual rather than literal. The Lovász extension keeps the finite ground set fixed and extends a set function from indicator vectors to arbitrary real vectors. Graphon theory instead enlarges the ambient space and studies limits of increasingly large graphs. Still, both are examples of a recurring analytic move in modern combinatorics:

discrete 0–1 data	$\longrightarrow$	real-valued analytic data
$\longrightarrow$	integration, limits, and geometry	

In graphon theory this passage supports graph-limit, regularity, sampling, and property-testing methods; in the present chapter it supports Choquet integration, convexity, support functions, and polyhedral optimization.

Let $f : 2^{[n]} \to \mathbb{R}$ with $f(\varnothing) = 0$. Given $x \in \mathbb{R}^n$, choose an ordering with $x_{\pi(1)} \geq x_{\pi(2)} \geq \cdots \geq x_{\pi(n)}$, and set $S_i := \{\pi(1), \ldots, \pi(i)\}$. The *Lovász extension* of f is

$$\widehat{f}(x) = x_{\pi(n)} f([n]) + \sum_{i=1}^{n-1} (x_{\pi(i)} - x_{\pi(i+1)}) f(S_i).$$

Equivalently,

$$\widehat{f}(x) = \sum_{i=1}^{n} x_{\pi(i)} (f(S_i) - f(S_{i-1})).$$

It agrees with f on 0–1 vectors: $\widehat{f}(\mathbf{1}_X) = f(X)$.

Theorem 1.13 (Lovász)**.** *A set function f is submodular if and only if its Lovász extension $\widehat{f}$ is convex.*

We shall prove this after introducing the associated polyhedron.

1.4 Submodular and base polyhedra

For $x \in \mathbb{R}^U$ and $X \subseteq U$, write $x(X) = \sum_{u \in X} x_u$. Define the submodular polyhedron by $P(f) := \{x \in \mathbb{R}^U : x(X) \leq f(X) \text{ for every } X \subseteq U\}$, and define its *base polyhedron* by $B(f) := \{x \in P(f) : x(U) = f(U)\}$.

Theorem 1.14 (Greedy description of a base polyhedron)**.** *Assume f is submodular and $f(\varnothing) = 0$. For every ordering $\pi(1), \dots, \pi(n)$ define $y_{\pi(i)} = f(S_i) - f(S_{i-1})$. Then $y \in B(f)$. Moreover, for any weight vector c sorted so that $c_{\pi(1)} \geq \cdots \geq c_{\pi(n)}$, this greedy vector maximizes $c^T x$ over $B(f)$.*

Proof. For any $X = \{\pi(i_1), \dots, \pi(i_t)\}$ with $i_1 < \cdots < i_t$, diminishing returns gives

$$y_{\pi(i_j)} = f(S_{i_j}) - f(S_{i_j - 1}) \leq f(X \cap S_{i_j}) - f(X \cap S_{i_j - 1}).$$

Summing yields $y(X) \leq f(X)$, while $y(U) = f(U)$. Hence $y \in B(f)$. For optimality, summation by parts gives

$$c^T x = c_{\pi(n)} x(U) + \sum_{i=1}^{n-1} (c_{\pi(i)} - c_{\pi(i+1)}) x(S_i) \leq c_{\pi(n)} f(U) + \sum_{i=1}^{n-1} (c_{\pi(i)} - c_{\pi(i+1)}) f(S_i),$$

with equality for the greedy y. □

Corollary 1.15 (Vertices and integrality of base polyhedra)**.** *Every vertex of $B(f)$ is a greedy vector associated with some ordering of U. Consequently, if f is integer-valued, then every vertex of $B(f)$ is integral.*

Proof. Let v be a vertex. Choose a linear objective $c^T x$ uniquely maximized at v, and perturb c slightly so that its coordinates are distinct while keeping v as the unique maximizer. Let π order the coordinates of c decreasingly. Theorem 1.14 supplies a greedy vector y maximizing the same objective. Uniqueness forces $y = v$. If f is integer-valued, every coordinate $y_{\pi(i)} = f(S_i) - f(S_{i-1})$ is an integer. □

Remark 1.16 (Subgradients of the Lovász extension). On every cone $x_{\pi(1)} \geq \cdots \geq x_{\pi(n)}$, the Lovász extension is linear with coefficient vector equal to the greedy base vector for π. Thus the greedy construction simultaneously describes vertices of $B(f)$ and subgradients of the convex function $\widehat{f}$. This is one precise sense in which "submodularity is discrete convexity."

Proof of Theorem 1.13. Assume first that f is submodular. Fix $x \in \mathbb{R}^n$ and choose an ordering π with $x_{\pi(1)} \geq \cdots \geq x_{\pi(n)}$. Apply Theorem 1.14 with the weight vector $c = x$. The greedy vector $y^\pi \in B(f)$ maximizes $y^T x$ over $B(f)$, and the value of that maximum is

$$\begin{aligned}(y^\pi)^T x &= x_{\pi(n)} f([n]) + \sum_{i=1}^{n-1} (x_{\pi(i)} - x_{\pi(i+1)}) f(S_i) \\ &= \widehat{f}(x).\end{aligned}$$

Consequently

$$\widehat{f}(x) = \max_{y \in B(f)} y^T x = h_{B(f)}(x). \tag{8}$$

Thus the Lovász extension is the support function of the base polyhedron. By the epigraph observation preceding Theorem 1.12, every support function is convex: it is a pointwise supremum of linear functions. This proves the forward implication without repeating the same maximum-of-linear-functions calculation.

Conversely, suppose that $\widehat{f}$ is convex. Let $X, Y \subseteq [n]$, put $A := X \cap Y$ and $B := X \cup Y$, and let $q := (\mathbf{1}_X + \mathbf{1}_Y)/2$. By convexity, applied with $\lambda = 1/2$, and because $\widehat{f}(\mathbf{1}_Z) = f(Z)$ for every $Z \subseteq [n]$,

$$\widehat{f}(q) \leq \frac{\widehat{f}(\mathbf{1}_X) + \widehat{f}(\mathbf{1}_Y)}{2} = \frac{f(X) + f(Y)}{2}. \tag{9}$$

Now q has value 1 on A, value $1/2$ on $B \setminus A = X \triangle Y$, and value 0 outside B. Thus, in the decreasing ordering used in the Lovász-extension formula, the coordinates occur in three blocks:

$$A \quad (\text{level } 1), \qquad B \setminus A \quad (\text{level } 1/2), \qquad [n] \setminus B \quad (\text{level } 0).$$

Only the two drops in level contribute. Hence

$$\begin{aligned} \widehat{f}(q) &= (1 - \tfrac{1}{2}) f(A) + (\tfrac{1}{2} - 0) f(B) \\ &= \frac{1}{2} f(X \cap Y) + \frac{1}{2} f(X \cup Y). \end{aligned}$$

Combining this equality with (9) and multiplying by 2 gives $f(X \cap Y) + f(X \cup Y) \leq f(X) + f(Y)$, which is precisely the submodular inequality. Therefore f is submodular. □

Proposition 1.17 (Level-set formula for the Lovász extension)**.** *Let $f : 2^{[n]} \to \mathbb{R}$ satisfy $f(\varnothing) = 0$, and for $x \in \mathbb{R}^n$ put $S_t(x) := \{i \in [n] : x_i \geq t\}$. Then*

$$\widehat{f}(x) = \int_0^\infty f(S_t(x))\, dt + \int_{-\infty}^0 (f(S_t(x)) - f([n]))\, dt. \tag{10}$$

In particular, if $x \geq 0$ coordinatewise, then

$$\widehat{f}(x) = \int_0^\infty f(S_t(x))\, dt. \tag{11}$$

If moreover $f([n]) = 0$, then for arbitrary $x \in \mathbb{R}^n$,

$$\widehat{f}(x) = \int_{-\infty}^\infty f(S_t(x))\, dt. \tag{12}$$

Thus the Lovász extension is literally assembled from the values of the original set function on the nested superlevel sets of x.

Proof. Suppose first that $x \geq 0$, and order its coordinates as $x_{\pi(1)} \geq \cdots \geq x_{\pi(n)} \geq 0$. Put $S_i = \{\pi(1), \ldots, \pi(i)\}$ and $x_{\pi(n+1)} := 0$. Whenever $x_{\pi(i+1)} < t \leq x_{\pi(i)}$, the superlevel set $S_t(x)$ is exactly S_i (endpoints do not affect the integral). Hence

$$\int_0^\infty f(S_t(x))\, dt = \sum_{i=1}^n (x_{\pi(i)} - x_{\pi(i+1)}) f(S_i),$$

which is precisely the defining formula for $\widehat{f}(x)$.

For general x, choose $a \geq -\min_i x_i$, so that $z = x + a\mathbf{1} \geq 0$. The definition of the Lovász extension immediately gives $\widehat{f}(x + a\mathbf{1}) = \widehat{f}(x) + a f([n])$. Applying the nonnegative formula to z and changing variables $s = t - a$ gives

$$\widehat{f}(x) = \int_{-a}^\infty f(S_s(x))\, ds - a f([n]).$$

Splitting at 0 yields (10); for sufficiently negative s we have $S_s(x) = [n]$, so the second integrand is eventually zero and the lower limit may be written as $-\infty$. The two stated special cases follow immediately. □

The Choquet-integral viewpoint

The preceding proposition is exactly the finite-space formula for the Choquet integral. We record the identification explicitly because it connects submodular optimization with a much older theory of capacities.

Definition (Capacity and finite Choquet integral)**.** A *capacity* on a finite ground set U is a monotone set function $\nu : 2^U \to \mathbb{R}$ with $\nu(\varnothing) = 0$. In probabilistic applications one often also normalizes $\nu(U) = 1$. For $x \in \mathbb{R}^U$ its (asymmetric) Choquet integral is

$$C_\nu(x) := \int_0^\infty \nu(\{u : x_u \geq t\})\, dt + \int_{-\infty}^0 (\nu(\{u : x_u \geq t\}) - \nu(U))\, dt. \tag{13}$$

For $x \geq 0$ only the first integral is present.

Proposition 1.18 (Lovász extension equals the Choquet integral)**.** *If ν is a capacity on U, then*

$$\widehat{\nu}(x) = C_\nu(x) \qquad (x \in \mathbb{R}^U).$$

If ν is additive, say $\nu(A) = \sum_{u \in A} p_u$, then

$$C_\nu(x) = \sum_{u \in U} p_u x_u,$$

so the Choquet integral reduces to the ordinary integral on the finite measure space (U, ν).

Proof. The first assertion is exactly Proposition 1.17. For the additive case, either substitute into (13) and interchange the finite sum with the integral, or use the second formula in the definition of the Lovász extension: every marginal difference is simply the weight $p_{\pi(i)}$. □

The word *capacity* comes from potential theory. Choquet's foundational paper developed a general theory of increasing set functions and classical capacities, with strong connections to analytic sets and potential theory [7]. The integral now bearing his name became a central tool in nonadditive measure theory; a systematic modern treatment is Denneberg [10]. It also entered probability and statistics in several ways. For example, the hitting probabilities

$$T(K) = \Pr\{Z \cap K \neq \varnothing\}$$

of a random closed set Z form a capacity functional and, under the appropriate continuity and alternating conditions, determine the law of Z; see Molchanov [46]. In models of ambiguity and nonadditive probability, a capacity replaces a single additive probability and the Choquet integral plays the role of expectation. Schmeidler's representation theorem characterizes the Choquet integral by, among other properties, additivity on comonotone functions [57]; his later work made Choquet expected utility a basic model of decision under ambiguity [58].

The finite-dimensional formula also makes the key surviving form of additivity completely transparent.

Proposition 1.19 (Comonotone additivity)**.** *Call $x, z \in \mathbb{R}^U$* comonotone *if*

$$(x_u - x_v)(z_u - z_v) \geq 0 \qquad (u, v \in U),$$

so that the two vectors admit a common decreasing ordering of their coordinates. For every normalized set function f,

$$\widehat{f}(x + z) = \widehat{f}(x) + \widehat{f}(z)$$

whenever x and z are comonotone. Also $\widehat{f}(\alpha x) = \alpha \widehat{f}(x)$ for $\alpha \geq 0$ and $\widehat{f}(x + a\mathbf{1}) = \widehat{f}(x) + a f(U)$.

Proof. Choose one ordering π that is decreasing for both x and z. It is then decreasing for $x+z$ as well. On the corresponding order cone the Lovász extension is the same linear form

$$y^\pi \cdot w, \qquad y^\pi_{\pi(i)} = f(S_i) - f(S_{i-1}),$$

for $w = x, z, x+z$. Additivity follows. Positive homogeneity and translation along $\mathbf{1}$ follow directly from the defining formula. □

Thus ordinary integration is fully additive because an additive measure makes the extension linear on all of $\mathbb{R}^U$; Choquet integration retains exact additivity only along directions that preserve the same level-set chain. This is precisely why orderings, chains, and braid cones appear so naturally in the convex geometry developed below.

Example (Matroid rank: Choquet integration becomes greedy optimization)**.** Let M be a matroid on U with rank function r_M. Since r_M is a polymatroid rank function, its Lovász extension is its Choquet integral. If $x_{\pi(1)} \geq \cdots \geq x_{\pi(n)}$, then each rank increment

$$r_M(S_i) - r_M(S_{i-1})$$

is either 0 or 1. The positions where it equals 1 are exactly the elements selected by the matroid greedy algorithm. Consequently, if B_x is the greedy maximum-weight basis,

$$\widehat{r}_M(x) = \sum_{e \in B_x} x_e = \max\{x(B) : B \text{ is a basis of } M\}. \tag{14}$$

For $x \geq 0$, the layer-cake formula becomes

$$\widehat{r}_M(x) = \int_0^\infty r_M(\{e : x_e \geq t\})\, dt.$$

At every threshold, the greedy basis contains a basis of the surviving set, so the integral literally accumulates "how many independent elements remain above level t."

For the uniform matroid $U_{k,n}$ this says

$$\widehat{r}(x) = x_{(1)} + \cdots + x_{(k)},$$

the sum of the k largest coordinates. For the graphic matroid of a connected graph, (14) is the value of a maximum-weight spanning tree. In this sense Kruskal's algorithm computes a Choquet integral by lowering a threshold through the edge weights and recording the rank increases when components merge.

1.5 Graph cuts become total variation

The convex extension of a graph cut has a particularly beautiful form.

Theorem 1.20 (Cut function as graph total variation)**.** *Let $G = (V, E)$ have nonnegative edge weights w_e, and define $c(X) := \sum_{uv \in \delta(X)} w_{uv}$. Then its Lovász extension is $\widehat{c}(x) = \sum_{uv \in E} w_{uv} |x_u - x_v|$.*

Proof. Both sides are additive over edges, so it suffices to consider one edge uv. The corresponding set function is 1 exactly when a set separates u and v. Its Lovász extension is $|x_u - x_v|$. □

A genuine bridge between discrete and continuous mathematics: why "coarea"? For $t \in \mathbb{R}$, write $S_t := \{v \in V : x_v \geq t\}$. For a single edge uv one has the elementary identity

$$|x_u - x_v| = \int_{-\infty}^{\infty} |\mathbf{1}_{\{u \in S_t\}} - \mathbf{1}_{\{v \in S_t\}}|\, dt. \tag{15}$$

Indeed, if $x_u \geq x_v$, then the integrand is 1 exactly for $x_v < t \leq x_u$ and is 0 otherwise, so its

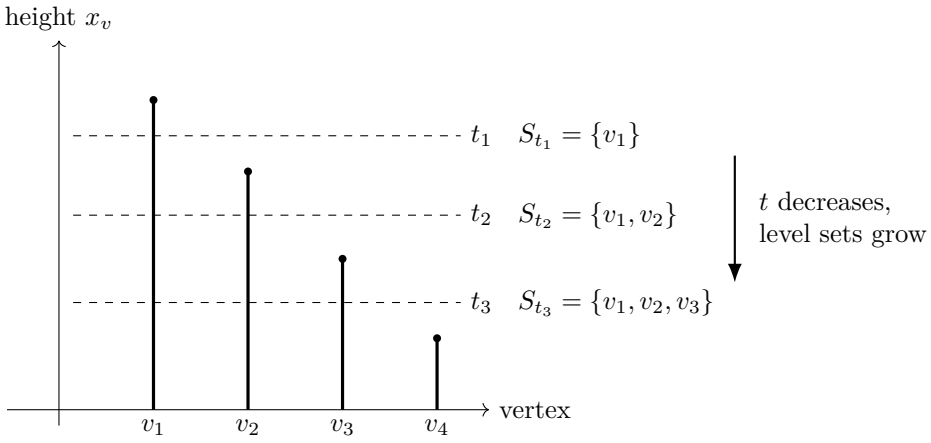


Figure 2: A real vector x determines a nested family of superlevel sets $S_t = \{v : x_v \geq t\}$. The Lovász extension weights these discrete sets by the lengths of the height intervals for which they persist.

integral is $x_u - x_v$. Multiplying (15) by w_{uv} and summing over the edges gives

$$\underbrace{\sum_{uv \in E} w_{uv}|x_u - x_v|}_{\text{graph total variation of } x} = \int_{-\infty}^{\infty} \underbrace{c(S_t)}_{\text{weighted boundary of the level set } S_t} dt. \tag{16}$$

This is called a *coarea identity* because it is the discrete graph analogue of the coarea formula for functions of bounded variation in analysis, where the total variation of a function is obtained by integrating the perimeters of its superlevel sets; see, for example, Evans–Gariepy [17, Chapters 3 and 5]. On a graph, the cut capacity $c(S)$ plays exactly the role of the perimeter of S.

There is a second, purely submodular reason for the same identity. A cut function satisfies $c(\varnothing) = c(V) = 0$, so Proposition 1.17 gives

$$\widehat{c}(x) = \int_{-\infty}^{\infty} c(S_t)\, dt,$$

whereas Theorem 1.20 gives $\widehat{c}(x) = \sum_{uv} w_{uv}|x_u - x_v|$. Thus the graph coarea formula is not an isolated trick: it is the level-set representation of the Lovász extension specialized to cuts.

How this connects to the rest of the notes. As t decreases, the sets S_t are nested. If $x_{\pi(1)} \geq \cdots \geq x_{\pi(n)}$, then, between successive coordinate values, these superlevel sets are exactly the chain

$$\varnothing \subset S_1 \subset S_2 \subset \cdots \subset S_n = V$$

used in the definition of the Lovász extension and in the greedy description of the base polyhedron. Thus a continuous vector x is encoded by a chain of discrete sets together with the lengths of the intervals on which each set is a superlevel set. This is also why chain and laminar structures keep reappearing in the uncrossing and polyhedral parts of the notes: the level-set picture is a canonical source of chains, although it should not itself be confused with the uncrossing argument.

If $x \in [0,1]^V$, then (16) reduces to

$$\sum_{uv} w_{uv}|x_u - x_v| = \int_0^1 c(S_t)\,dt.$$

The right-hand side is the *average* cost of the threshold cuts S_t. Consequently at least one threshold has cost no larger than the continuous total-variation value. This is the threshold-rounding principle used again in the later subsection on coarea and cuts. The same mechanism underlies many Cheeger-type and clustering arguments: one first constructs a real-valued function and then extracts a discrete set by choosing a good level.

BRIDGE: cuts $\longleftrightarrow$ total variation and coarea. The graph-theoretic object $c(S)$ and the convex function $\widehat{c}(x) = \sum_{uv} w_{uv}|x_u - x_v|$ are not merely analogous: the latter is obtained by integrating the former over the threshold sets of x. This is the first explicit meeting of the two reading paths.

For a directed arc $u \to v$, the Lovász extension of its in-cut contribution is $(x_v - x_u)_+ := \max\{x_v - x_u, 0\}$. Hence $\widehat{\rho_D}(x) = \sum_{uv \in A}(x_v - x_u)_+$. There is a directed coarea identity as well. Since

$$(x_v - x_u)_+ = \int_{-\infty}^{\infty} \mathbf{1}_{\{x_u < t \le x_v\}}\,dt,$$

summing over the arcs gives

$$\widehat{\rho_D}(x) = \int_{-\infty}^{\infty} \rho_D(S_t)\,dt.$$

Thus the one-sided total variation is an average of directed in-cuts of the same nested level sets. This is a useful mental model for the later directed connectivity and orientation theorems: continuous one-sided variation and discrete directed cuts are two descriptions of the same level-set geometry.

2 Uncrossing as Convex Geometry

The elementary uncrossing lemma from the preceding section has a much richer interpretation than its two-line proof might suggest. In this chapter we develop that interpretation systematically. There are three pictures of the same phenomenon:

1. *lattice picture:* tight subsets are stable under union and intersection;
2. *polyhedral picture:* tight subsets are equations defining faces of a base polyhedron;
3. *normal-fan picture:* chains of tight subsets record the directions from which those faces are exposed.

For graph cut functions a fourth picture appears: the base polyhedron is a graphical zonotope, so cut inequalities, bounded flows, total variation, and convex geometry become different descriptions of one object.

This section takes a deliberately geometric viewpoint. The basic graph-theoretic uncrossing arguments later in the handout can be followed without all of this machinery, but the convex picture explains why those arguments are so reusable and why many integral graph theorems are shadows of convex objects with unusually rigid faces.

2.1 Tight sets are the equations of a face

Let $f : 2^U \to \mathbb{R}$ be submodular with $f(\varnothing) = 0$, and let $B(f) = \{x \in \mathbb{R}^U : x(S) \le f(S)\ (S \subseteq U),\ x(U) = f(U)\}$. For $x \in B(f)$, write $\mathcal{T}(x) = \{S \subseteq U : x(S) = f(S)\}$ for the family of sets

whose inequalities are tight at x.

Every vector $x \in \mathbb{R}^U$ defines a modular set function

$$m_x(S) := x(S) = \sum_{u \in S} x_u.$$

Hence the base polyhedron has a useful functional interpretation:

$$B(f) = \{x : m_x(S) \leq f(S) \text{ for all } S \subseteq U,\ m_x(U) = f(U)\}. \tag{17}$$

In words, $B(f)$ is the family of modular functions lying below f and agreeing with f on the whole ground set. There is an exact convex-analytic version of this statement.

Proposition 2.1 (Modular minorants are supporting linear functions)**.** *For normalized submodular f,*

$$B(f) = \{b \in \mathbb{R}^U : b^T x \leq \widehat{f}(x) \text{ for every } x \in \mathbb{R}^U\} = \partial \widehat{f}(0), \tag{18}$$

where $\partial \widehat{f}(0)$ denotes the set of vectors b satisfying $\widehat{f}(x) \geq \widehat{f}(0) + b^T x$ for every x. Consequently the hyperplane $t = b^T x$ supports the epigraph of $\widehat{f}$ from below. For a subset $S \subseteq U$,

$$b(S) = f(S) \quad \Longleftrightarrow \quad b^T \mathbf{1}_S = \widehat{f}(\mathbf{1}_S),$$

so a tight set is exactly a 0–1 *point at which this supporting linear function touches the Lovász extension.*

Proof. If $b \in B(f)$, then the support-function formula (8) gives $b^T x \leq \widehat{f}(x)$ for every x. Conversely, suppose this inequality holds for every x. Taking $x = \mathbf{1}_S$ gives $b(S) \leq f(S)$ for every S. Taking $x = \mathbf{1}_U$ and $x = -\mathbf{1}_U$, and using $\widehat{f}(a\mathbf{1}) = af(U)$ for all $a \in \mathbb{R}$, gives $b(U) = f(U)$. Hence $b \in B(f)$. Since $\widehat{f}(0) = 0$, the first set in (18) is precisely the subdifferential $\partial \widehat{f}(0)$. The final equivalence follows from $\widehat{f}(\mathbf{1}_S) = f(S)$. □

Thus a tight set is a *contact set* where a modular minorant touches the submodular function, and at the same time an indicator vector where a supporting hyperplane touches the convex epigraph of the Lovász extension. Faces of $B(f)$ record common patterns of such contacts. This is not merely an analogy: the modular, polyhedral, and convex-function pictures are literally the same supporting geometry viewed in three coordinate systems.

Proposition 2.2 (The tight-set lattice)**.** *For every $x \in B(f)$, the family $\mathcal{T}(x)$ is closed under union and intersection.*

Proof. Take $X, Y \in \mathcal{T}(x)$. Modularity of $S \mapsto x(S)$ gives $f(X) + f(Y) = x(X) + x(Y) = x(X \cap Y) + x(X \cup Y)$. Since $x \in B(f)$,

$$x(X \cap Y) + x(X \cup Y) \leq f(X \cap Y) + f(X \cup Y) \leq f(X) + f(Y),$$

where the final inequality is submodularity. Equality holds throughout, so both $X \cap Y$ and $X \cup Y$ are tight. □

Thus a point of $B(f)$ does not merely satisfy a list of active inequalities: its active subsets form a lattice. This is already a convex-geometric statement, because active inequalities are precisely the supporting hyperplanes containing the smallest face through the point.

The ordinary separating-hyperplane theorem also has an unusually discrete form for a submodular base polyhedron.

Proposition 2.3 (A separating hyperplane collapses to a violated subset)**.** *Let f be normalized and submodular, and let $p \in \mathbb{R}^U$ satisfy $p(U) = f(U)$. Then the following are equivalent:*

(i) $p \in B(f)$;

(ii) $c^T p \le \widehat{f}(c)$ *for every* $c \in \mathbb{R}^U$;

(iii) $p(S) \le f(S)$ *for every* $S \subseteq U$.

Moreover, suppose $p \notin B(f)$ *and a vector* c *strictly separates* p *from* $B(f)$*:*

$$c^T p > h_{B(f)}(c) = \widehat{f}(c).$$

Order the coordinates so that $c_{\pi(1)} \ge \cdots \ge c_{\pi(n)}$ *and put* $S_i = \{\pi(1), \ldots, \pi(i)\}$*. Then at least one level set* S_i *across a strict drop of* c *satisfies*

$$p(S_i) > f(S_i).$$

Thus an arbitrary real separating direction can be replaced by the 0–1 *direction* $\mathbf{1}_{S_i}$ *of an actually violated subset inequality.*

Proof. The equivalence of (i) and (iii) is the definition of $B(f)$ under the assumed equality $p(U) = f(U)$. The implication (i)⇒(ii) follows from the support-function formula. Conversely, (ii) applied to $c = \mathbf{1}_S$ gives (iii).

For the final assertion, summation by parts gives

$$c^T p = c_{\pi(n)} p(U) + \sum_{i=1}^{n-1} (c_{\pi(i)} - c_{\pi(i+1)}) p(S_i),$$

whereas the Lovász formula gives

$$\widehat{f}(c) = c_{\pi(n)} f(U) + \sum_{i=1}^{n-1} (c_{\pi(i)} - c_{\pi(i+1)}) f(S_i).$$

The total-set terms cancel because $p(U) = f(U)$. All remaining coefficients are nonnegative. If the first display is strictly larger than the second, at least one term with positive coefficient must satisfy $p(S_i) > f(S_i)$. □

A discrete separation principle. For a general convex body, a separating normal vector may point in an arbitrary real direction. For a submodular base polyhedron, thresholding that direction exposes a violated subset. In this sense the exponentially many 0–1 normals $\mathbf{1}_S$ are already a complete language of separation. This is one reason submodular polyhedra admit such concrete combinatorial certificates.

The next result says that, although all tight sets may form a complicated distributive lattice, a single *chain* of them is enough to describe any exposed face.

Theorem 2.4 (Chains expose faces)**.** *Let* $c \in \mathbb{R}^U$ *take the distinct values* $\beta_1 > \beta_2 > \cdots > \beta_m$*. Put* $C_j = \{u : c_u \ge \beta_j\}$ *for* $j = 1, \ldots, m$*; thus*

$$\varnothing \subsetneq C_1 \subsetneq C_2 \subsetneq \cdots \subsetneq C_m = U.$$

Then the face of $B(f)$ *maximizing* $c^T x$ *is exactly* $\{x \in B(f) : x(C_j) = f(C_j),\ j = 1, \ldots, m-1\}$*. Consequently every face of* $B(f)$ *can be exposed by requiring a chain of subsets to be tight.*

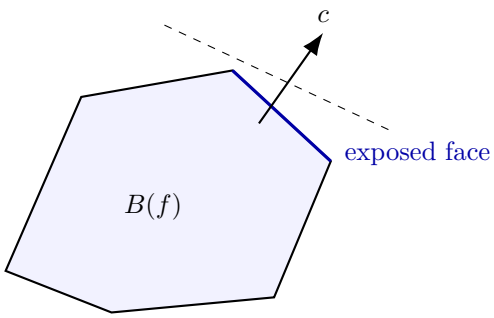
Proof. Summation by parts gives, for every $x \in B(f)$,

$$c^T x = \beta_m f(U) + \sum_{j=1}^{m-1} (\beta_j - \beta_{j+1}) x(C_j).$$

Since every coefficient $\beta_j - \beta_{j+1}$ is positive and $x(C_j) \le f(C_j)$,

$$c^T x \le \beta_m f(U) + \sum_{j=1}^{m-1} (\beta_j - \beta_{j+1}) f(C_j),$$

with equality precisely when all the displayed chain inequalities are tight. Every face of a polytope is exposed by some linear functional c, so the final assertion follows. □



maximizing $c^T x$ pushes the polytope against a supporting line

Figure 3: An exposed face of a base polyhedron is selected by a linear functional. Theorem 2.4 says that, for submodular base polyhedra, the active equations needed to describe such a face can be chosen from a nested chain of sets.

Uncrossing as face compression. The full tight-set lattice may contain exponentially many members, but a face can be reached from a direction c by remembering only the nested threshold sets of c. Thus uncrossing is not only a way to simplify a proof: it reflects the fact that the normal geometry of a submodular polyhedron is organized by chains.

2.2 Normal cones, the braid fan, and greedy orderings

For a polytope $P \subseteq \mathbb{R}^U$ and a face F of P, define its *normal cone*

$$N_P(F) := \{c \in \mathbb{R}^U : F \subseteq \arg\max_{x \in P} c^T x\}.$$

The relative interior of $N_P(F)$ consists of the directions that expose *exactly* the face F. The collection of all normal cones, one for each face, is the *normal fan* of P. Thus the normal fan is a decomposition of objective space: it records which face is optimal for each linear objective.

For a base polytope, Theorem 2.4 says more. If c lies in a normal cone, only the chain of superlevel sets of c matters. Crossing subsets never have to be remembered independently. In particular, the walls at which the greedy answer can change are controlled by equalities of coordinates.

The *braid arrangement* in $\mathbb{R}^U$ consists of the hyperplanes $c_u = c_v$ for $u \ne v$. Its chambers are the regions

$$c_{u_1} > c_{u_2} > \cdots > c_{u_n},$$

so they are indexed by total orderings of U. The greedy base vector is constant in each braid chamber, although several adjacent chambers may lead to the same face or even the same vertex.

Theorem 2.5 (Base polytopes are generalized permutahedra)**.** *The normal fan of a submodular base polytope $B(f)$ is a coarsening of the braid fan. Equivalently, every edge of $B(f)$ is parallel to a vector $e_u - e_v$.*

Geometric proof sketch. Inside one chamber of the braid arrangement the order of the coordinates of c is fixed, hence the greedy maximizer is fixed. Therefore each braid chamber lies inside a normal cone of $B(f)$, which says precisely that the normal fan of $B(f)$ coarsens the braid fan. The equivalent edge-direction description follows by polarity between codimension-one walls of the normal fan and edge directions. Polytopes with this property are called generalized permutahedra; see, for example, Postnikov [53]. □

This normal-fan picture is also the geometric form of comonotone additivity from Proposition 1.19: two comonotone vectors lie in a common braid cone, and on that cone the Lovász/Choquet extension is one fixed linear functional. Thus the same ordering simultaneously controls the integral formula, the greedy algorithm, and the face exposed in the base polytope.

2.3 Greedy vertices, integrality, and generalized permutahedra

If c has pairwise distinct coordinates, the chain in Theorem 2.4 is maximal. The greedy theorem from the preceding section then gives a unique maximizer.

Corollary 2.6 (Every vertex is greedy)**.** *Every vertex of $B(f)$ is obtained from an ordering $u_1, \dots, u_n$ by $x_{u_i} = f(\{u_1, \dots, u_i\}) - f(\{u_1, \dots, u_{i-1}\})$. In particular, if f is integer-valued, then $B(f)$ is an integral polytope.*

Proof. Expose a given vertex by a vector c in the relative interior of its normal cone and perturb c slightly so that its coordinates are distinct without changing the exposed vertex. The greedy theorem identifies the unique maximizer. If f is integer-valued, every greedy difference is integral. □

This one observation contains a large amount of combinatorial integrality. For example, if r is the rank function of a matroid M on U, then $B(r)$ is the convex hull of the incidence vectors of the bases of M. Indeed, the greedy differences of a matroid rank function are 0 or 1, and the positions where they equal 1 form the basis selected by the matroid greedy algorithm.

Example 2.7 (The ordinary permutahedron)**.** Let $a_1 \geq a_2 \geq \cdots \geq a_n$ and define $f(S) = a_1 + \cdots + a_{|S|}$. The decreasing increments make f submodular. The greedy vertices of $B(f)$ are precisely the permutations of $(a_1, \dots, a_n)$, so $B(f)$ is the usual permutahedron generated by that vector.

Example 2.8 (Matroid base polytopes)**.** For a matroid M, the generalized permutahedron $B(r_M)$ is $\operatorname{conv}\{\mathbf{1}_B : B \text{ is a basis of } M\}$. For the graphic matroid of a connected graph, its vertices are the incidence vectors of spanning trees. Thus the spanning-tree polytope is itself a generalized permutahedron, now living in edge-space.

At the *matroid level*, every greedy increment is 0 or 1, so the vertices above are incidence vectors. For an integral polymatroid the same greedy construction still gives integral vertices, but coordinates may be larger than 1. For a completely general integral submodular function, the base polytope is still integral by Theorem 2.6, even though the function need not be monotone at all. Thus matroid base polytopes, polymatroid base polytopes, and the cut base polytopes below are successive specializations of the same greedy base-polyhedral geometry.

The last example already foreshadows a recurring theme: a spanning tree may appear as a *vertex* of one convex body and, as we shall soon see, as a *volume contribution* to another.

2.4 Minkowski addition of base polytopes

Submodular functions add, and their convex bodies add as well.

Proposition 2.9 (Minkowski additivity)**.** *If f and g are normalized submodular functions on the same ground set, then*

$$B(f+g) = B(f) + B(g),$$

where the right-hand side is the Minkowski sum.

Proof. The Lovász extension is additive:

$$\widehat{f+g}(c) = \widehat{f}(c) + \widehat{g}(c).$$

By Theorem 1.14, these three functions are respectively the support functions of $B(f+g)$, $B(f)$, and $B(g)$. The support function of a Minkowski sum is the sum of the support functions, and compact convex sets are determined by their support functions. □

This is especially effective for graphs because cut functions are sums of one-edge cut functions.

2.5 Graph cuts are graphical zonotopes

For $v \in V$, let e_v denote the corresponding coordinate vector in $\mathbb{R}^V$.

Consider first one directed arc $a = u \to v$ of capacity $w \geq 0$, with set function

$$f_a(X) = \begin{cases} w, & u \notin X,\ v \in X, \\ 0, & \text{otherwise.} \end{cases}$$

Its Lovász extension is $w(x_v - x_u)_+$. But this is also the support function of the segment $[0, w(e_v - e_u)]$. Hence $B(f_a) = [0, w(e_v - e_u)]$. For an undirected edge $e = uv$ of weight w, the one-edge cut function has Lovász extension $w|x_v - x_u|$, the support function of the centered segment $[-w(e_v - e_u),\ w(e_v - e_u)]$. Using Theorem 2.9 edge by edge gives the following concrete description.

Theorem 2.10 (Cut base polytopes are zonotopes)**.** *Let $D = (V, A)$ be a capacitated digraph, with capacity w_a on arc $a = u \to v$. Then*

$$B(\rho_D) = \sum_{a=u\to v \in A} [0, w_a(e_v - e_u)].$$

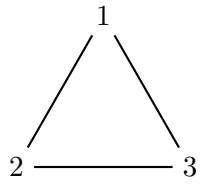
For an undirected weighted graph $G = (V, E)$,

$$B(d_G) = \sum_{e=uv \in E} [-w_e(e_v - e_u),\ w_e(e_v - e_u)].$$

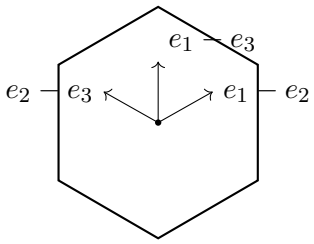
Consequently the support functions of these zonotopes are respectively

$$\widehat{\rho_D}(x) = \sum_{u\to v} w_{uv}(x_v - x_u)_+, \qquad \widehat{d_G}(x) = \sum_{uv} w_{uv}|x_v - x_u|.$$

A Minkowski sum of line segments is a *zonotope*. Thus the familiar cut function does not merely possess some abstract base polyhedron: its base polyhedron is one of the classical objects of convex geometry. This is deliberately a step *beyond polymatroids*: cut functions are generally nonmonotone, so their zonotopes belong to the general submodular base-polyhedron theory rather than to polymatroid rank theory.



triangle graph K_3



a two-dimensional graphical zonotope

The picture is schematic: after quotienting by the all-ones direction, three edge directions in the triangle generate a centrally symmetric hexagon.

One graph, three convex objects. For a connected graph G there are at least three useful convex bodies nearby.

1. The spanning-tree polytope $B(r_{M(G)}) \subseteq \mathbb{R}^E$ has spanning trees as vertices.
2. The cut base polytope $B(d_G) \subseteq \mathbb{R}^V$ is a graphical zonotope and has graph total variation as support function.
3. Flow polytopes live in edge-space, but their images under the incidence matrix are precisely the zonotopes in (2).

Changing which space we look at changes a graph theorem from a statement about vertices, to one about support functions, to one about linear images.

2.6 Euclidean projection and submodular minimization

The support-function picture optimizes a *linear* functional over $B(f)$. A second geometric operation, Euclidean projection, unexpectedly solves a discrete minimization problem. This is one of the cleanest demonstrations that the base polyhedron is not merely a visualization device.

Because U is finite, $B(f)$ is a nonempty compact convex polytope. Hence there is a unique point

$$b^* := \operatorname*{arg\,min}_{b \in B(f)} \|b\|_2.$$

Put

$$S^- := \{u : b^*_u < 0\}, \qquad S^0 := \{u : b^*_u \le 0\}.$$

GEOMETRIC PROOF — nearest point + a supporting face. The signs of the point of $B(f)$ nearest the origin recover the extremal minimizers of f. The proof uses only the first-order geometry of Euclidean projection together with the chain description of exposed faces.

Theorem 2.11 (Fujishige's minimum-norm-base principle)**.** *Let $f : 2^U \to \mathbb{R}$ be submodular with $f(\varnothing) = 0$, and let b^* be the minimum-norm point of $B(f)$. Then S^- and S^0 both minimize f. More precisely, every minimizer S of f satisfies*

$$S^- \subseteq S \subseteq S^0. \tag{19}$$

Consequently S^- is the unique inclusion-minimal minimizer and S^0 is the unique inclusion-maximal minimizer.

Proof. The projection characterization of the nearest point says

$$\langle b^*, b - b^* \rangle \ge 0 \qquad (b \in B(f)).$$

Equivalently, b^* maximizes the linear functional $(-b^*)^T b$ over $B(f)$. Apply Theorem 2.4 to the exposing vector $c = -b^*$. Its positive superlevel set is S^-, and its nonnegative superlevel set is S^0; allowing the trivial cases $\varnothing$ and U, both are tight at b^*. Hence

$$f(S^-) = b^*(S^-) = \sum_{u:b_u^*<0} b_u^* = b^*(S^0) = f(S^0).$$

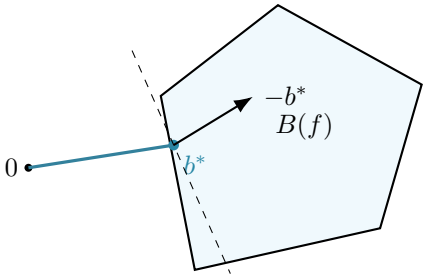
For an arbitrary $S \subseteq U$,

$$f(S) \geq b^*(S) \geq \sum_{u:b_u^*<0} b_u^*.$$

The second inequality holds because omitting a negative coordinate can only increase the sum, while including a positive coordinate can only increase it. Thus S^- and S^0 attain the global minimum.

If S is any minimizer, equality must hold in the second inequality. Therefore every negative coordinate belongs to S and no positive coordinate belongs to S; zero coordinates are the only freedom. This is exactly (19). □

This theorem is due to Fujishige's minimum-norm/lexicographically optimal base theory; see Fujishige [26, 27] and the algorithmic development by Fujishige–Isotani [28]. The theorem is a structural reduction, not by itself a claim that Euclidean projection is computationally cheaper than submodular minimization; the minimum-norm-point algorithms exploit the fact that linear optimization over $B(f)$ is available by the greedy theorem. Notice how tightly the result fits the earlier modular-minorant picture. The point b^* defines a modular function below f, and the minimizer is found precisely where that modular minorant touches f while collecting all of its negative coordinates.



the supporting line at the nearest point is perpendicular to b^*

Figure 4: The minimum-norm point b^* is the Euclidean projection of the origin onto $B(f)$. The vector $-b^*$ exposes the supporting face through b^*; its threshold sets are tight, and the zero threshold identifies the minimizers of f.

Example (A graph optimization problem as projection onto a translated zonotope)**.** Let $G = (V, E)$ be an undirected graph with nonnegative edge weights, let $c(S)$ be its cut capacity, and let $a \in \mathbb{R}^V$ be vertex prizes or penalties. Consider

$$F(S) := c(S) + a(S). \tag{20}$$

The first term is submodular and the second is modular, so F is submodular. Moreover a modular summand simply translates a base polytope: by Proposition 2.9, since the base polytope of $S \mapsto a(S)$ is the singleton $\{a\}$,

$$B(F) = B(c) + a.$$

By Theorem 2.10, $B(c)$ is a graphical zonotope. Therefore minimizing the discrete objective (20) is geometrically equivalent to the following operation:

translate the graphical zonotope by a, project 0 onto it, and threshold the signs.

If b^* is the nearest point of $B(c) + a$ to the origin, then $\{v : b^*_v < 0\}$ is the unique minimal optimal vertex set and $\{v : b^*_v \leq 0\}$ is the unique maximal one.

Ordinary s–t minimum cut is contained in this picture. Let $W = \sum_{e \in E} w_e$ and choose $M > W$. With

$$a = M(e_s - e_t),$$

we have

$$F(S) = c(S) + M\mathbf{1}_{\{s \in S\}} - M\mathbf{1}_{\{t \in S\}}.$$

Every minimizer must contain t and omit s: such a set has value $c(S) - M < 0$, whereas every set failing one of these terminal conditions has value at least 0. Among the admissible sets, minimizing F is exactly minimizing the s–t cut capacity. Thus an s–t minimum cut can also be read from the sign pattern of the point nearest the origin in the translated graphical zonotope

$$B(c) + M(e_s - e_t).$$

This is conceptually different from the radial-intersection proof below: there we move along one prescribed direction until we hit the zonotope; here we translate the whole zonotope and ask which point is closest to the origin.

Two geometric operations, two discrete optimization principles. The same cut zonotope now supports two distinct pictures.

1. A *radial intersection* in the terminal direction $e_t - e_s$ gives the maximum-flow/minimum-cut value.
2. A *Euclidean projection* after a modular translation gives the minimizer of a cut-plus-vertex-cost objective.

Supporting hyperplanes explain both. In the first, a separating functional is rounded by level sets; in the second, the normal vector at the nearest point is the point itself, and its sign threshold gives the discrete optimizer. This is a concrete reason to regard modular functions, supporting faces, and submodularity as parts of one geometric mechanism rather than separate technical devices.

2.7 Flows as points of a cut base polytope

The zonotope description makes max-flow/min-cut almost visible. Choose an arbitrary orientation of the edges of an undirected graph and let A be the vertex-edge incidence matrix whose column for $u \to v$ is $e_v - e_u$. A vector $z \in \mathbb{R}^E$ represents signed flow on the oriented edges, and Az is the vector of net inflow at the vertices.

Proposition 2.12 (Cut base polytopes are imbalance polytopes)**.** *For an undirected graph with capacities w_e,*

$$B(d_G) = \{Az : -w_e \leq z_e \leq w_e \text{ for all } e\}.$$

For a capacitated digraph, with the incidence matrix oriented as the arcs,

$$B(\rho_D) = \{Az : 0 \leq z_a \leq w_a \text{ for all } a\}.$$

Proof. The image under A of the box $\prod_e [-w_e, w_e]$ is the Minkowski sum of the segments $[-w_e A_e, w_e A_e]$, where A_e is the corresponding column. Apply Theorem 2.10. The directed case is identical with intervals $[0, w_a]$. □

Thus membership in a cut base polytope asks whether a prescribed vector of supplies and demands can be realized by a bounded flow.

Theorem 2.13 (Max-flow/min-cut as the radius of a zonotope)**.** *Let D be a capacitated digraph and let $s \neq t$. The maximum value of an s–t flow equals*

$$\max\{\lambda \geq 0 : \lambda(e_t - e_s) \in B(\rho_D)\}.$$

Moreover,

$$\max\{\lambda \geq 0 : \lambda(e_t - e_s) \in B(\rho_D)\} = \min\{\rho_D(X) : t \in X,\ s \notin X\}.$$

Hence max-flow/min-cut says that the radial extent of the directed cut zonotope in the direction $e_t - e_s$ is exactly the minimum s–t cut value.

Proof. By Theorem 2.12, the membership $\lambda(e_t - e_s) \in B(\rho_D)$ is equivalent to the existence of a capacity-feasible arc vector whose net inflow is $-\lambda$ at s, $+\lambda$ at t, and 0 elsewhere; this is exactly an s–t flow of value λ.

On the other hand, by the defining inequalities of the base polytope,

$$\lambda(e_t - e_s)(X) \leq \rho_D(X) \qquad (X \subseteq V).$$

The left side equals λ precisely when $t \in X$ and $s \notin X$, equals $-\lambda$ in the opposite case, and equals 0 otherwise. Therefore the only upper bounds on positive λ are

$$\lambda \leq \rho_D(X) \qquad (t \in X,\ s \notin X),$$

and the largest feasible value is the minimum of these numbers. □

GEOMETRIC PROOF — separation + coarea. There is a second proof of the min–max equality that uses the geometry of the zonotope directly and makes the earlier coarea formula do real work.

Second proof of the min–cut equality. Write $Z = B(\rho_D) = \sum_{a=u\to v}[0, w_a(e_v - e_u)]$ and put $q = e_t - e_s$. Let $\mu = \min\{\rho_D(X) : t \in X,\ s \notin X\}$. The cut inequalities already give the easy direction: if $\lambda q \in Z$ and X is an s–t cut side, then $\lambda = (\lambda q)(X) \leq \rho_D(X)$, so $\lambda \leq \mu$.

For the reverse inequality apply the point–convex-set separation theorem, Theorem 1.12, to the compact zonotope Z. Suppose, for a contradiction, that $\mu q \notin Z$. Then there is $y \in \mathbb{R}^V$ with

$$\mu(y_t - y_s) > h_Z(y),$$

where the support function of the directed cut zonotope is $h_Z(y) = \sum_{a=u\to v} w_a(y_v - y_u)_+$. Because $0 \in Z$, the displayed strict inequality forces $y_t - y_s > 0$. Adding a constant to all coordinates of y and then scaling by the positive number $y_t - y_s$, we may assume $y_s = 0$ and $y_t = 1$.

For $0 < \theta < 1$, set $S_\theta = \{v : y_v \geq \theta\}$. Then $t \in S_\theta$ and $s \notin S_\theta$, so every such level set is an s–t cut side. For a single arc $u \to v$, the set of thresholds for which that arc enters S_θ has length at most $(y_v - y_u)_+$. Summing over the arcs gives

$$\int_0^1 \rho_D(S_\theta)\, d\theta \leq \sum_{a=u\to v} w_a(y_v - y_u)_+ = h_Z(y) < \mu.$$

But every S_θ is an s–t cut side, hence $\rho_D(S_\theta) \geq \mu$ for all $0 < \theta < 1$, and the same integral is at least μ, a contradiction. Therefore $\mu q \in Z$, which proves the reverse inequality. □

This is quite literally the separating-hyperplane theorem translated into graph theory. The excluded point is the desired imbalance vector $\mu(e_t - e_s)$; a separating normal y certifies that it does not lie in the flow zonotope; and the support value $h_Z(y)$ is the Lovász/Choquet extension of the directed cut function. Thresholding y converts the continuous separator into 0–1 cut

directions. The coarea/layer-cake identity says that the support value is an average of the costs of those threshold cuts, so one of them would have to beat the alleged minimum cut—the contradiction.

Compare this with Proposition 2.3. There an arbitrary separator of a base polyhedron collapses directly to a violated subset inequality. Here the same philosophy is used in a more informative way: instead of merely invoking the H-description of $B(\rho_D)$, we start from the *zonotope/image-of-a-capacity-box* description and recover a cut certificate by separation plus threshold integration. This is why the argument is genuinely geometric rather than just a restatement of the defining cut inequalities.

Classroom experiment. Take a small directed network with four or five vertices. Compute the image of its capacity box under the incidence matrix, then intersect the resulting zonotope with the line $\{\lambda(e_t - e_s) : \lambda \in \mathbb{R}\}$. The endpoint of the intersection is the maximum flow value. Ask students to identify, from the face supporting that endpoint, a minimum cut.

2.8 Spanning trees appear again: volume of the graphical zonotope

There is a second, rather surprising way in which spanning trees enter the convex geometry of cuts. Choose an orientation of each edge of a connected graph G and define the translated graphical zonotope

$$Z_G = \sum_{e=u\to v\in E} [0, e_v - e_u] \subseteq H := \left\{ x \in \mathbb{R}^V : \sum_{v\in V} x_v = 0 \right\}.$$

The lattice $H \cap \mathbb{Z}^V$ has rank $|V| - 1$. We measure volume using this lattice, so a fundamental parallelepiped has normalized volume 1.

Theorem 2.14 (Spanning trees as zonotope volume)**.** *For a connected unweighted graph G,*

$$\mathrm{Vol}_{\mathrm{norm}}(Z_G) = \tau(G),$$

where $\tau(G)$ is the number of spanning trees of G. More generally, replacing the segment for edge e by $[0, w_e(e_v - e_u)]$ gives normalized volume

$$\sum_{T \text{ spanning tree}} \prod_{e\in T} w_e.$$

Proof sketch. A basic fact about zonotopes says that the volume of a Minkowski sum of segments is the sum of the absolute determinants of all maximal independent sets of generators. Here a set of $|V| - 1$ incidence vectors $e_v - e_u$ is linearly independent in H exactly when the corresponding edges form a spanning tree. For a tree, deleting one row from its incidence matrix gives a square matrix of determinant ± 1. Thus every spanning tree contributes normalized volume 1, and dependent edge sets contribute 0. Scaling the segment of e by w_e multiplies the corresponding determinant contribution by w_e. □

The weighted formula is a convex-geometric incarnation of the spanning-tree polynomial and sits naturally beside the matrix-tree theorem.

The same spanning trees twice. Spanning trees are the *vertices* of the graphic matroid base polytope in $\mathbb{R}^E$, while the same spanning trees index the elementary parallelepipeds whose volumes add up to the volume of the graphical zonotope in $\mathbb{R}^V$. One appearance is controlled by matroid rank submodularity; the other by the incidence vectors underlying graph cuts.

Exercise 2.15. For a connected graph G on n vertices, show that the centered cut base polytope $B(d_G)$ is a translate of $2Z_G$. Deduce from Theorem 2.14 that $\mathrm{Vol}_{\mathrm{norm}}\, B(d_G) = 2^{n-1}\tau(G)$ for an unweighted graph.

Exercise 2.16. Use the Cauchy–Binet formula to derive the weighted matrix-tree theorem from the determinant contributions in the proof of Theorem 2.14.

2.9 Uncrossing in linear-programming duals

The face geometry above concerns the subset inequalities $x(S) \le f(S)$ of a base polytope. A second form of uncrossing occurs in dual linear programs whose variables themselves are indexed by cuts. This is the mechanism behind many laminar dual certificates in network design.

Let $a_X \in \mathbb{R}^m_{\ge 0}$ be a constraint vector associated with $X \subseteq V$. Consider the covering LP

$$\min\{c^T x : a_X^T x \ge p(X) \text{ for all } X,\ x \ge 0\}$$

and its dual

$$\max\left\{\sum_X p(X) y_X : \sum_X y_X a_X \le c,\ y_X \ge 0\right\}.$$

Suppose

$$a_X + a_Y \ge a_{X\cap Y} + a_{X\cup Y} \tag{UC}$$

coordinatewise and

$$p(X) + p(Y) \le p(X \cap Y) + p(X \cup Y). \tag{SM}$$

If a dual solution has positive weights on crossing X, Y, put $\varepsilon = \min\{y_X, y_Y\}$ and move ε units of weight from X, Y to $X \cap Y, X \cup Y$. Condition (UC) says dual feasibility is preserved, while (SM) says the objective does not decrease.

The standard cut vector $a_X = \mathbf{1}_{\delta(X)}$ satisfies the coordinatewise inequality (UC). Indeed, every edge can be checked separately. By symmetry it also satisfies the posimodular companion

$$\mathbf{1}_{\delta(X)} + \mathbf{1}_{\delta(Y)} \ge \mathbf{1}_{\delta(X\setminus Y)} + \mathbf{1}_{\delta(Y\setminus X)}.$$

This leads to the following common generalization.

Definition 2.17 (Skew-supermodularity)**.** A set function $p : 2^V \to \mathbb{R}$ is *skew-supermodular* if for every $X, Y \subseteq V$ at least one of $p(X) + p(Y) \le p(X \cap Y) + p(X \cup Y)$ or $p(X) + p(Y) \le p(X \setminus Y) + p(Y \setminus X)$ holds.

For such requirements, one can uncross a crossing pair using whichever of the two alternatives is valid. A secondary potential makes the simplification quantitative. For example, under union-intersection uncrossing use the potential $\Phi(y) := \sum_X y_X |X|\,|V \setminus X|$. If X, Y cross and $b := |X \setminus Y|$, $c := |Y \setminus X|$, then replacing one unit on X, Y by one unit on $X \cap Y, X \cup Y$ changes this potential by $-2bc < 0$. Thus repeated uncrossing cannot cycle if the data are rational and a suitable optimal solution with minimal potential is selected.

Proposition 2.18 (Laminar dual principle)**.** *In a cut covering problem for which the requirement and the allowed domain support the uncrossing operation above, an optimal dual solution can be chosen with laminar support.*

The exact hypotheses vary from theorem to theorem: sometimes one uses ordinary supermodularity, sometimes crossing supermodularity, and sometimes skew-supermodularity together with symmetry. The important point is the proof pattern: crossing dual mass can be moved to simpler sets without harming feasibility or value.

Primal faces versus dual laminarity. There are two distinct uncrossing mechanisms that are easy to conflate.

1. In a base polytope, submodularity of the right-hand side forces the *active primal inequalities* to form a lattice.
2. In a cut covering LP, coordinatewise submodularity of cut incidence vectors together with supermodularity of the requirement lets us move *dual weight* from crossing sets to uncrossed sets.

Both ultimately compress exponentially indexed subset information into a nested certificate, but the algebraic directions of the inequalities are different.

2.10 Iterative rounding: convex geometry becomes approximation

A striking modern application of the preceding principle is Jain's factor-2 approximation for survivable network design [36]. One version starts from pairwise connectivity requirements $r_{uv} \in \mathbb{Z}_{\geq 0}$ and asks for a minimum-cost subgraph having at least r_{uv} edge-disjoint u–v paths for every pair. The natural relaxation has cut constraints

$$x(\delta(S)) \geq \max\{r_{uv} : u \in S,\ v \notin S\} \qquad (\varnothing \neq S \subsetneq V), \quad x \geq 0.$$

The residual cut requirement arising during the algorithm has the appropriate skew-supermodular structure.

The key geometric statement is not that every extreme point is integral; that is false. Instead one proves a *large-coordinate lemma*: at every nonzero basic feasible solution of the relevant residual LP, some edge satisfies $x_e \geq \frac{1}{2}$. The proof first uncrosses tight cut constraints until a laminar family spans all tight constraints needed to determine the basic solution. One then performs a token-counting argument on the laminar forest. If all coordinates were strictly smaller than $1/2$, there would not be enough independent tight equations to determine the point.

Rounding such an edge up to 1, updating the residual requirement, and repeating produces a solution of cost at most twice the optimum fractional cost.

Why this belongs in a course on submodularity. The factor 2 does not come from a probabilistic rounding estimate. It comes from the geometry of an extreme point: uncrossing says that a basis of tight constraints can be organized by a laminar forest, and the forest is sparse enough to force one large coordinate. This is an unusually vivid example of a structural graph theorem controlling an approximation ratio.

Project 2.19 (Reconstruct the token argument)**.** Study Jain's large-coordinate lemma in the special case of a uniform requirement k. Starting from a basic feasible solution $0 < x_e < 1/2$, uncross its tight cuts to a laminar family and try to assign two tokens per edge to the nodes of the laminar forest so that every tight set receives enough tokens and at least one token remains. Identify exactly where integrality of the right-hand side is used.

Project 2.20 (Partition and copartition uncrossing)**.** In orientation and augmentation problems, ordinary set uncrossing may fail and the correct dual objects become partitions and copartitions. Investigate the uncrossing theory used in minimum-cost connectivity orientation and augmentation. The broader question is: *what is the right convex-geometric substitute for laminar families when the basic constraints are indexed by partitions rather than subsets?*

2.11 Minimum cuts: lattices, ideals, and cactus geometry

For a fixed ordered pair s, t, minimum s–t cut sides containing t and excluding s form a lattice under union and intersection. There is an even sharper description. After one maximum flow

is found, contract the strongly connected components of the residual digraph. The resulting acyclic graph encodes all minimum cuts: the minimum t-sides correspond to suitable ideals (or, depending on the orientation convention, filters) of this DAG. Thus a family of exponentially many minimum cuts is represented by a finite poset.

This is an instructive endpoint of ordinary uncrossing: a distributive lattice of optimal cuts becomes a lattice of ideals.

For *global* minimum cuts in an undirected graph, however, crossing cannot always be eliminated. Cyclic crossing patterns genuinely occur. The correct structure theorem is therefore not a laminar tree but a cactus.

Theorem 2.21 (Cactus representation of global minimum cuts)**.** *Every undirected weighted graph admits a cactus representation of all its global minimum cuts: there is a cactus C together with a map from $V(G)$ to $V(C)$ such that the bipartitions represented by minimum cuts of the cactus are exactly the global minimum-cut bipartitions of G.*

Here a cactus is a connected graph in which every edge belongs to at most one simple cycle. The theorem originates in work of Dinitz, Karzanov, and Lomonosov [12]. In a cactus, the surviving nonlaminar behavior is completely controlled by cycles. This makes a useful slogan:

uncrossing removes arbitrary overlap; cactus cycles record the irreducible overlap.

The cactus theorem is also algorithmically alive. He, Huang, and Saranurak gave fast constructions of cactus representations and extended the methodology to Steiner-cactus and hypercactus representations [33, 34]. Thus the Dinitz–Karzanov–Lomonosov structural theorem of 1976 [12] remains part of current algorithmic graph theory.

Project 2.22 (From lattices to cacti)**.** Construct examples of graphs whose global minimum cuts contain crossing pairs that cannot all be made laminar. Compute a cactus representation by hand. Compare this with the fixed-s, t case, where the minimum cut sides do form a distributive lattice. What extra symmetry in the global problem forces cycles rather than arbitrary crossing patterns?

Research question 2.23 (A general "cactus principle")*.* For which symmetric submodular functions do all global minimizers admit a cactus-like representation? Graph cut functions have more structure than arbitrary symmetric submodular functions because they are sums of two-point cut functions. Can one characterize the additional algebraic or polyhedral property responsible for cactus representability?

2.12 Uncrossing, matroids, and Hall revisited

Before moving to the direct proof of Hall's theorem, it is useful to see matching through one more convex lens. Two different submodular functions naturally appear. First, the neighborhood function

$$p(X) := |N(X)| \qquad (X \subseteq L)$$

is normalized, monotone, integer-valued, and submodular, so it is an *integral polymatroid rank function*. It is usually not a matroid rank function: a single vertex u may have $p(\{u\}) = \deg(u) > 1$.

Second, call $I \subseteq L$ *matchable* if there is a matching saturating I. The matchable subsets of L form a transversal matroid. Its rank function is

$$r(X) := \max\{|I| : I \subseteq X \text{ is matchable}\},$$

which is a genuine matroid rank function. Hall's inequalities $p(X) \geq |X|$ compare the polymatroidal coverage function p with the modular cardinality function, and Hall's conclusion says

exactly that the matroid rank reaches its maximum value $r(L) = |L|$. Thus Hall's theorem is a particularly clean meeting point of the polymatroid and matroid levels.

The base polytope of the transversal matroid is integral by Theorem 2.6. Hence Hall's theorem, matroid greedy integrality, and matching all meet in the same convex framework, although the elementary uncrossing proof in the next section is far shorter than this conceptual route.

BRIDGE: Hall $\longleftrightarrow$ transversal-matroid geometry. The direct proof later uses tight neighborhood sets and induction; the convex route packages the same matchability structure into a submodular rank function and an integral base polytope. The handout deliberately presents both viewpoints.

2.13 Research directions suggested by the convex picture

The following problems are deliberately broad enough to serve as seminar projects or seeds for research.

Project 2.24 (The normal fan under graph operations)**.** For a graph cut function d_G, study how the normal fan of $B(d_G)$ changes under deletion, contraction, splitting-off, and pinching. Deletion removes one zonotope generator; contraction identifies two coordinates. Splitting-off is subtler. Can admissibility of a splitting operation be recognized as a local statement about faces or normal cones of the associated zonotope?

Research question 2.25 (A convex splitting-off theorem)*.* The usual proofs of splitting-off are driven by dangerous sets and uncrossing. Is there a formulation in which complete splitting-off at a vertex becomes a decomposition or projection theorem for a cut base polyhedron? More ambitiously, can one recover Lovász–Mader splitting-off from a statement about preserving selected radial functions of a zonotope?

Project 2.26 (Quantitative uncrossing)**.** Classical uncrossing usually proves existence of a laminar optimum. Study the number of elementary uncrossing moves required to transform an arbitrary rational optimum into a laminar one. Can strongly polynomial bounds be obtained for natural graph LPs? Which potentials give the sharpest estimates?

Project 2.27 (Beyond submodularity)**.** Bisubmodular and k-submodular functions have convex extensions and polyhedral theories analogous to ordinary submodularity. Find graph problems whose natural tight objects are signed sets or multi-label sets, and ask whether their uncrossing arguments are better understood through these generalized forms of discrete convexity.

Research question 2.28 (Zonotopes and connectivity certificates)*.* The support function of $B(d_G)$ records total variation, while the radial function in a terminal direction records a max-flow value. Which other classical graph parameters can be read naturally as widths, radii, face dimensions, mixed volumes, or lattice-point counts of $B(d_G)$ or related base polytopes?

Project 2.29 (Cactus, Steiner cactus, and hypercactus)**.** Read the Dinitz–Karzanov–Lomonosov cactus theorem [12] together with the algorithms of He, Huang, and Saranurak for cactus, Steiner-cactus, and hypercactus representations [33, 34]. Identify which parts of the proofs are pure uncrossing and which use specifically graphical structure. A particularly attractive direction is to search for a common submodular formulation that explains why all three representations remain sparse.

2.14 Exercises on uncrossing as geometry

Exercise 2.30 (Tight-set distributive lattice)**.** Let $x \in B(f)$. Prove that $\mathcal{T}(x)$, ordered by inclusion, is a distributive lattice. Give an example for which $\mathcal{T}(x)$ is not laminar.

Exercise 2.31 (Choquet integral and ordinary integration)**.** Let ν be a capacity on U. Prove Proposition 1.19 directly from the level-set formula. Then show that C_ν is linear on all of $\mathbb{R}^U$ if and only if ν is modular. Interpret the conclusion as the exact point at which the Choquet integral collapses to an ordinary finite measure integral.

Exercise 2.32 (Uniform matroid and top-k sums)**.** For the uniform matroid $U_{k,n}$, derive directly from the Choquet integral that $\widehat{r}(x) = x_{(1)} + \cdots + x_{(k)}$. Draw the unit ball of the dual piecewise-linear functional in the first nontrivial low-dimensional cases and identify the braid-fan walls where the maximizing k-set changes.

Exercise 2.33 (Minimum norm and a translated cut zonotope)**.** Take a weighted path or cycle on four vertices, choose a nonzero vector $a \in \mathbb{R}^V$, and write down the translated zonotope $B(c) + a$. Compute its minimum-norm point and verify Theorem 2.11 by comparing its sign pattern with all values of $c(S) + a(S)$.

Exercise 2.34 (Face dimension from blocks)**.** Let c have level sets $B_1, \ldots, B_m$ and chain $C_j = B_1 \cup \cdots \cup B_j$. Under suitable nondegeneracy assumptions, determine the dimension of the face exposed by c in terms of how f decomposes across the blocks. Work this out explicitly for a matroid base polytope.

Exercise 2.35 (A directed one-arc base polytope)**.** Starting only from the inequalities defining $B(f_a)$ for one arc $a = u \to v$, prove directly that $B(f_a) = [0, w(e_v - e_u)]$. Then recover the directed zonotope formula without using support functions.

Exercise 2.36 (Minimum cut as supporting face)**.** Let $b = \lambda(e_t - e_s)$ be the endpoint of $B(\rho_D) \cap \{\mu(e_t - e_s) : \mu \geq 0\}$. Show that every minimum s–t cut X with $t \in X$ defines a supporting hyperplane $z(X) = \rho_D(X)$ containing b. Interpret the lattice of minimum cuts as part of the normal cone structure at b.

Exercise 2.37 (Graphic matroid versus graphical zonotope)**.** For the cycle C_n, describe both the graphic matroid base polytope in $\mathbb{R}^{E(C_n)}$ and the graphical zonotope in $\mathbb{R}^{V(C_n)}/\langle \mathbf{1} \rangle$. Count the vertices of the first and compute the normalized volume of the second.

Exercise 2.38 (Uncrossing potential)**.** Verify the identity

$$|X \cap Y|\,|V \setminus (X \cap Y)| + |X \cup Y|\,|V \setminus (X \cup Y)| - |X|\,|V \setminus X| - |Y|\,|V \setminus Y| = -2|X \setminus Y|\,|Y \setminus X|.$$

Explain why this is a useful strict potential for union-intersection uncrossing.

Exercise 2.39 (A conjecture-making exercise)**.** Compute $B(d_G)$ for several small graphs G and tabulate easily visible geometric data: number of vertices, number of facets, normalized volume, and widths in root directions $e_v - e_u$. Guess which graph invariants these quantities encode. Prove at least one of your guesses and find a counterexample to another.

What to remember from this chapter. A submodular inequality is not merely a four-set estimate. It is the local algebra behind a global convex geometry. Tight sets are contact equations for modular minorants; chains expose faces; the normal fan is organized by braid cones; greedy differences are vertices; integer submodularity gives integral vertices; graph cut base polytopes are zonotopes; Euclidean projection can reveal minimizers; bounded flows are points of cut zonotopes; minimum cuts are supporting faces; and spanning trees reappear as normalized volume. Uncrossing is the combinatorial operation that lets us move between these descriptions without carrying the full exponential family of subsets.

Part II
Classical min–max theorems through the uncrossing lens

3 Hall's theorem as the first uncrossing proof

Throughout this section, graphs are finite. Let $G = (L, R; E)$ be a finite bipartite graph. Hall's condition is

$$|N(X)| \geq |X| \qquad (X \subseteq L).$$

The left side is submodular, the right side modular. Thus tight sets $|N(X)| = |X|$ uncross.

UNCROSSING CASE STUDY 1/8 — Hall (complete tight-set proof). This is the prototype. The induction itself needs only one proper tight set at a time; uncrossing explains why, when many tight sets are present, they form a structured family from which canonical minimal or maximal obstructions may be chosen. Thus "uncrossing" enters here as the structural principle behind the tight-set decomposition, even though the displayed induction does not literally replace a crossing pair.

Theorem 3.1 (Hall)**.** *Let $G = (L, R; E)$ be a finite bipartite graph. There is a matching saturating L if and only if $|N(X)| \geq |X|$ for every $X \subseteq L$.*

Proof. Necessity is immediate. For sufficiency, induct on $|L|$.

If there exists a nonempty proper tight set $X \subsetneq L$, then Hall's condition holds in the induced bipartite graph $G[X, N(X)]$, so by induction it has a matching saturating X. For $Y \subseteq L \setminus X$,

$$|N(Y) \setminus N(X)| = |N(X \cup Y)| - |N(X)| \geq |X \cup Y| - |X| = |Y|.$$

Thus Hall's condition holds between $L \setminus X$ and $R \setminus N(X)$, and induction completes the matching.

Now suppose there is no nonempty proper tight set. Choose any edge uv with $u \in L$. Delete u and v. For any nonempty $X \subseteq L \setminus \{u\}$ we have $|N_G(X)| \geq |X| + 1$, because X is not tight. Deleting v reduces its neighborhood by at most one, so Hall's condition survives. Induction gives a matching in the smaller graph; add uv. □

Remark (Why finiteness is being stated explicitly). The theorem above is the finite form of Hall's theorem, and the proof just given is genuinely finite: it is an induction on $|L|$. There are important infinite versions of Hall-type matching theorems, but they require additional ideas and should not be inferred from this induction. In this handout, every use of Hall's theorem is in the finite setting.

The usual existence statement is the zero-deficiency case of a genuine min–max formula. Write $\nu(G)$ for the maximum size of a matching and define the *left deficiency* by $\operatorname{def}_L(G) := \max_{X \subseteq L}(|X| - |N(X)|)$. Since $X = \varnothing$ is allowed, the deficiency is always nonnegative.

Theorem 3.2 (Hall's min–max form, or deficiency formula)**.** *For every finite bipartite graph $G = (L, R; E)$,*

$$\nu(G) = \min_{X \subseteq L}\big(|L \setminus X| + |N(X)|\big) = |L| - \operatorname{def}_L(G).$$

Equivalently, the minimum possible number of unmatched vertices of L in a matching is exactly $\operatorname{def}_L(G)$. In particular, Hall's condition is precisely the case $\operatorname{def}_L(G) = 0$.

Proof. Fix $X \subseteq L$. In any matching, at most $|N(X)|$ vertices of X can be matched, while at most $|L \setminus X|$ matched vertices can lie outside X. Hence $\nu(G) \leq |L \setminus X| + |N(X)|$ for every X.

For the reverse inequality, let $d := \mathrm{def}_L(G)$ and adjoin d new vertices to the right side, each adjacent to every vertex of L. In the enlarged graph, every nonempty $X \subseteq L$ has $|N(X)|+d \geq |X|$ neighbors, while the empty set causes no condition. Hall's theorem therefore gives a matching saturating L. After deleting the edges incident with the d new right vertices, at most d matched edges disappear, leaving a matching of size at least $|L| - d$ in the original graph. Together with the preceding upper bound, this proves the formula. □

Remark (Why this is a min–max theorem). The maximum on the packing side is the number $\nu(G)$ of pairwise disjoint matching edges. The minimum on the obstruction side is taken over subsets $X \subseteq L$ and charges one unit for every left vertex outside X plus one unit for every available neighbor of X. Hall's familiar condition asks only whether this common optimum reaches the extreme value $|L|$; the deficiency formula records the optimum even when a perfect saturation of L is impossible.

Hall's theorem as a prototype. The proof above already contains a recurring architecture of modern combinatorial optimization:

1. encode feasibility by inequalities coming from a submodular set function;
2. identify the tight sets, that is, the equality cases;
3. uncross tight sets until the obstruction has a simple nested, laminar, or lattice structure;
4. use that structure either to decompose the instance or to certify that a local move is safe.

The later theorems do not repeat these four steps verbatim, but they can be read as increasingly sophisticated versions of this scheme.

Where does the Hall architecture reappear? A forward map.

Minimum cuts and Menger; see Section 4.
The feasibility objects are s–t cut inequalities. Equality cases are minimum cuts, and submodularity makes them a lattice under union and intersection. The residual graph then singles out canonical extremal members of that lattice. Here the main payoff of uncrossing is *organization of optimal cuts*, rather than a local modification of the graph.

Edmonds' arborescences; see Section 5.
The inequalities are the in-cut conditions $\rho_D(X) \geq k$. Tight in-cuts are the obstructions to extending a partial arborescence. Minimal tight sets uncross, and one chooses an entering arc that avoids all tight obstructions. This is very close to Hall's "tight-set or safe local move" dichotomy.

Frank's free-root branchings; see Section 6.
Surviving auxiliary root arcs are certified by tight sets. Uncrossing merges intersecting certificates into a subpartition, and the subpartition inequality bounds how many root arcs can survive. This is a higher-dimensional version of replacing an arbitrary family of Hall obstructions by a structured one.

Tree packing; see Section 7.
The relevant obstructions are no longer single subsets but partitions. Matroid union converts the packing problem into a min–max rank formula, and the partition inequali-

ties play the role of the Hall inequalities. This section is therefore a nearby relative of the Hall architecture rather than a literal repetition of it.

Orientations and splitting-off; see Sections 8 and 9.
Desired connectivity is again expressed by cut lower bounds. An inadmissible split is witnessed by a nearly tight "dangerous" set. Dangerous sets uncross; a maximal one cannot contain every possible partner; hence a safe pair exists. This is perhaps the closest later analogue of Hall's final step: *if no decomposing tight set blocks us, make a local move.* The same orientation theorem is revisited by convex integrality in Section 10 and by monotone edge flips in Section 11.

Submodular flows; see Section 13.
Here the preceding pattern is abstracted. Ordinary conservation constraints are replaced by inequalities indexed by all vertex subsets, with a crossing-submodular right-hand side. Uncrossing controls tight inequalities and integrality turns a feasible fractional system into a combinatorial object. Standard references are Edmonds–Giles [15], Fujishige [27], Schrijver [60], and Frank [22].

For the graph-theoretic uses of submodularity and uncrossing more broadly, Frank's surveys [21, 20] and his monograph [22] are especially useful guides. The point of the map above is not that every theorem has the same proof, but that the same *proof vocabulary*—feasibility inequalities, tight sets, uncrossing, structured obstructions, and safe moves—keeps returning.

Remark (A different "marriage" theorem). Hall's theorem has a particularly instructive companion that does *not* belong to the same uncrossing template. Gale–Shapley stable marriage adds preference orders and replaces deficient sets by the local notion of a blocking pair. Its deferred-acceptance proof is driven by monotone proposals and rejections. The after-class project in Section 17 develops this comparison and then follows a 2022 slides to Galvin's stable-marriage proof of the Dinitz list-coloring theorem.

Exercise 3.3. For a bipartite graph define $f(X) = |N(X)| - |X|$. Prove directly that if $f \geq 0$ and X, Y are minimizers of f, then $X \cap Y$ and $X \cup Y$ are minimizers.

Exercise 3.4 (Weighted Hall)**.** Let $b : L \to \mathbb{N}$ and $c : R \to \mathbb{N}$. Find and prove the natural capacitated Hall condition for assigning $b(u)$ units of demand at each $u \in L$ to neighboring vertices of capacities $c(v)$. Identify the submodular function.

4 Menger, max-flow/min-cut, and the lattice of minimum cuts

Throughout this section, digraphs are finite. Let $D = (V, A)$ be a finite digraph and let $s, t \in V$ be *distinct* vertices. The directed edge form of Menger's theorem [45] says that the maximum number of pairwise arc-disjoint directed s–t paths equals the minimum size of an s–t cut.

Theorem 4.1 (Menger, directed edge form)**.** *Let $D = (V, A)$ be a finite digraph, let $s \neq t$, and let $k \in \mathbb{Z}_{\geq 0}$. The following are equivalent:*

(a) $\rho_D(X) \geq k$ for every X with $t \in X \subseteq V \setminus \{s\}$;

(b) there exist k pairwise arc-disjoint directed paths from s to t.

Proof. The implication (b)⇒(a) is immediate: every directed s–t path must use an arc entering any set X with $t \in X$ and $s \notin X$, and arc-disjoint paths use distinct such arcs.

For the converse, give every arc capacity 1 and run the elementary augmenting-path algorithm. A 0–1 flow is a function $f : A \to \{0, 1\}$ satisfying flow conservation at every vertex other than s and t. Its value is the net flow leaving s. Starting with the zero flow, form the residual digraph

D_f: an unused arc $u \to v$ appears in the forward direction, while a used arc $u \to v$ contributes the reverse residual arc $v \to u$. Whenever D_f contains a directed s–t path, toggle the value of f along that path: change 0 to 1 on forward residual arcs and change 1 to 0 on backward residual arcs. Conservation is preserved and the flow value increases by 1. Since the value is bounded by $|A|$, the procedure terminates.

Let S be the set of vertices reachable from s in the final residual digraph and put $X = V \setminus S$. Then $t \in X$ and $s \notin X$. No residual arc leaves S. Consequently every original arc from S to X is used by the flow, whereas every original arc from X to S is unused. Summing the conservation equations over the vertices of S therefore gives

$$|f| = f(\delta^+(S)) - f(\delta^-(S)) = |\delta^+(S)| = \rho_D(X).$$

Thus the final flow value equals the size of an s–t cut. Under hypothesis (a), this value is at least k.

Finally, an integral 0–1 flow of value q decomposes into q arc-disjoint directed s–t paths together with directed cycles: repeatedly follow positive-flow arcs from s until reaching t, remove that path, and then discard any remaining directed cycles. Hence the final flow contains at least k pairwise arc-disjoint s–t paths. □

Remark (Why s and t are required to be distinct). The statement above deliberately assumes $s \neq t$. With our cut convention, an s–t cut side is a set X with $t \in X$ and $s \notin X$. If $s = t$, no such set exists, so condition (a) becomes vacuous and cannot characterize a path packing. Allowing the trivial length-zero path from s to itself also makes the packing problem convention-dependent. If one instead asks for pairwise arc-disjoint *nontrivial* directed cycles through a specified vertex s, that is a different problem; it can be converted to a two-terminal path problem by splitting s into an "out-copy" and an "in-copy". We therefore keep the standard Menger convention $s \neq t$ throughout.

Remark (Why the proof is finite). The augmenting-path argument is also genuinely finite. Each augmentation raises the integral flow value by one, and in a finite digraph that value is bounded by $|A|$, so the algorithm terminates after finitely many augmentations. Infinite versions of Menger's theorem require substantially different arguments and are outside the scope of this handout.

The two-terminal formulation extends cleanly to two terminal sets. Let $S, T \subseteq V$ be nonempty and disjoint. An *S–T path* is a directed path whose first vertex lies in S and whose last vertex lies in T. An *S–T cut side* is a set X satisfying $T \subseteq X \subseteq V \setminus S$; its cut is the in-cut $\delta^-(X)$.

Corollary 4.2 (Menger between two terminal sets)**.** *Let $D = (V, A)$ be a finite digraph and let $S, T \subseteq V$ be nonempty and disjoint. For an integer $k \geq 0$, the following are equivalent:*

(a) $\rho_D(X) \geq k$ for every X with $T \subseteq X \subseteq V \setminus S$;

(b) there exist k pairwise arc-disjoint directed S–T paths.

Equivalently, the maximum number of pairwise arc-disjoint directed S–T paths equals $\min\{\rho_D(X) : T \subseteq X \subseteq V \setminus S\}$.

Proof. Add a new source s_0 and a new sink t_0. Put $M := |A| + 1$. For every $u \in S$, add M parallel arcs from s_0 to u, and for every $v \in T$, add M parallel arcs from v to t_0. Apply Theorem 4.1 to s_0, t_0 in the augmented finite digraph.

Any cut using one of the new bundles has size at least $M > |A|$, whereas deleting all original arcs gives an s_0–t_0 cut of size at most $|A|$. Hence every minimum s_0–t_0 cut leaves all of S on the source side and all of T on the sink side. Its original-arc part is therefore exactly $\delta_D^-(X)$ for some X with $T \subseteq X \subseteq V \setminus S$. Conversely, every such X gives an s_0–t_0 cut of the same size using only original arcs.

Likewise, deleting the first and last auxiliary arcs from a family of arc-disjoint s_0–t_0 paths gives arc-disjoint S–T paths. Conversely, every family of arc-disjoint S–T paths has at most $|A|$ members, since $S \cap T = \varnothing$ and every such path uses at least one original arc. It can therefore be extended through distinct auxiliary arcs because each bundle has $M > |A|$ parallel copies. The single-source theorem gives the asserted equality and equivalence. □

Remark (Why we assume $S \cap T = \varnothing$). If S and T overlap, length-zero terminal paths again enter the formulation and obscure the cut statement. For the clean edge-disjoint version used here, it is best to assume that S and T are nonempty and disjoint. Vertex-disjoint set-to-set versions are also standard, but their separator conventions require separate treatment.

Exercise 4.3 (Hall's deficiency theorem from set-to-set Menger)**.** Let $G = (L, R; E)$ be a finite bipartite graph. Construct a layered digraph D as follows. For each $u \in L$ introduce a private source vertex s_u, and for each $v \in R$ introduce a private sink vertex t_v. Put $S := \{s_u : u \in L\}$ and $T := \{t_v : v \in R\}$. Add the arcs $s_u \to u$ for $u \in L$, the arc $u \to v$ for every edge $uv \in E$, and the arcs $v \to t_v$ for $v \in R$.

(a) Show that a family of q pairwise arc-disjoint directed S–T paths in D is equivalent to a matching of size q in G.

(b) Apply Theorem 4.2 to obtain

$$\nu(G) = \min\{\rho_D(X) : T \subseteq X \subseteq V(D) \setminus S\},$$

where $\nu(G)$ is the maximum matching size.

(c) For an admissible cut side X, put $Y := L \setminus X$. Show that X may be replaced, without increasing $\rho_D(X)$, by the canonical cut side

$$X_Y := T \cup (L \setminus Y) \cup (R \setminus N(Y)).$$

Hint. If a vertex $v \in N(Y)$ lies in X, move it outside X: at least one arc from Y to v disappears from the in-cut, while only the arc $v \to t_v$ can be added. If $v \in R \setminus N(Y)$ lies outside X, move it into X.

(d) Compute $\rho_D(X_Y)$ and deduce the full Hall deficiency formula

$$\nu(G) = \min_{Y \subseteq L}\big(|L \setminus Y| + |N(Y)|\big) = |L| - \max_{Y \subseteq L}\big(|Y| - |N(Y)|\big).$$

Conclude Hall's theorem by asking when the common value equals $|L|$.

Remark (What the exercise shows). Hall's theorem appears earlier because its tight-set induction is the simplest model of uncrossing. The exercise reveals a complementary relation in the opposite direction: once set-to-set Menger is available, Hall's entire numerical min–max theorem is a special layered-network instance of path–cut duality. Thus Hall is pedagogically prior in the uncrossing story but mathematically contained in the broader flow framework.

Remark (Set-to-set Menger is still single-commodity; genuine multiflow is different). The preceding corollary may look like a multicommodity statement, but it is still a one-commodity problem: a path may start at any vertex of S and finish at any vertex of T, so a supersource and a supersink reduce the problem to one source–sink pair.

A genuine *multiflow* problem prescribes several terminal pairs (s_i, t_i), usually with separate demands, and asks that all commodities be routed simultaneously through the same capacities. There is no universal analogue of Menger's theorem in which one minimum cut by itself gives the optimum for arbitrary prescribed pairs. Even the most obvious integral cut condition is not sufficient. For example, let the supply graph be the four-cycle $v_1v_2v_3v_4v_1$ and ask for one path

joining v_1 to v_3 and one path joining v_2 to v_4. Every cut has at least as many supply edges as prescribed pairs separated by the cut, yet the two required paths cannot be chosen edge-disjoint.

For fractional multicommodity flow, linear-programming duality still gives a min–max theorem, but the dual object is naturally a system of edge lengths (equivalently, a metric-type certificate), not a single cut. Important special classes do admit sharper cut-type theorems; the point here is only that the simple one-pair Menger formula does not survive unchanged. See, for example, Schrijver [60] and Frank [22] for broader multiflow and disjoint-path theory.

Remark (Gomory–Hu trees: all pairwise cuts, but not a simultaneous multiflow). There is nevertheless a remarkable way to package *all pairwise single-commodity* minimum-cut values in an undirected capacitated graph. The Gomory–Hu theorem [31] constructs a weighted tree T on the same vertex set such that, for every pair u, v, the value of a minimum u–v cut in the original graph equals the minimum edge weight on the unique u–v path in T. Moreover, the fundamental cut determined by an appropriate minimum-weight tree edge on that path is a minimum u–v cut in the original graph.

Thus $n-1$ weighted tree edges encode the values of all $\binom{n}{2}$ pairwise minimum-cut problems. This should not be confused with a multiflow theorem: the tree tells us the optimum when one pair is considered at a time, but it does not assert that the corresponding pairwise flows can be routed simultaneously. Conceptually, the Gomory–Hu tree sits between ordinary Menger theory and genuine multiflow theory: it is a global compression of many single-commodity min–max theorems.

The proof above is the unit-capacity augmenting-path proof of max-flow/min-cut. Submodularity is not needed for the numerical equality; its additional role is to organize the whole family of minimum cuts.

UNCROSSING CASE STUDY 2/8 — the minimum-cut lattice (complete structural proof). Uncrossing is not needed for the numerical max-flow/min-cut equality. Its role is to show that the entire family of optimal cut sides is closed under union and intersection, producing canonical extremal cuts and, ultimately, a distributive-lattice description.

Proposition 4.4 (Minimum cuts form a lattice). *Suppose the minimum s–t cut has value λ. If X and Y are minimum s–t sides containing t but not s, then $X \cap Y$ and $X \cup Y$ are also minimum cut sides.*

Proof. Submodularity gives

$$2\lambda = \rho(X) + \rho(Y) \geq \rho(X \cap Y) + \rho(X \cup Y) \geq 2\lambda. \qquad \square$$

Thus there is a unique inclusion-minimal minimum t-side and a unique inclusion-maximal one. These extremal cuts also have a concrete max-flow description.

Proposition 4.5 (Canonical minimum cuts from a residual graph). *Let f be a maximum s–t flow in a capacitated digraph and let D_f be its residual digraph. Let*

$$R_s = \{v : v \text{ is reachable from } s \text{ in } D_f\}, \qquad R_t = \{v : t \text{ is reachable from } v \text{ in } D_f\}.$$

Then R_s is the unique inclusion-minimal source side of a minimum cut, while R_t is the unique inclusion-minimal t-side. Equivalently, $V \setminus R_s$ is the unique inclusion-maximal minimum t-side.

Proof. If S is the source side of any minimum cut for the maximum flow f, there is no residual arc from S to $V \setminus S$: every original arc leaving S is saturated and every original arc entering S carries zero flow across the cut. Hence every vertex reachable from s in D_f lies in S, so $R_s \subseteq S$ for every minimum source side S. Conversely, by definition no residual arc leaves R_s, and the usual max-flow/min-cut residual argument shows that $(R_s, V \setminus R_s)$ is a minimum cut. Thus R_s is the intersection of all minimum source sides.

The argument for R_t is dual. If X is any minimum t-side and a residual path from v to t started outside X, that path would contain a residual arc entering X, impossible for a minimum cut. Hence $R_t \subseteq X$ for every minimum t-side. No residual arc enters R_t, so $V \setminus R_t$ is a minimum source side and R_t itself is a minimum t-side. □

This is a first glimpse of the Picard–Queyranne description of the complete family of minimum cuts [52]: the lattice operation from submodularity is visible algorithmically in a single residual network.

Duality plus uncrossing. Max-flow/min-cut is a linear-programming duality theorem. Submodularity is not needed merely to prove the equality, but it explains why optimal dual solutions can be organized. This is a recurring pattern: LP duality gives an optimum; uncrossing gives a *combinatorial* optimum.

BRIDGE: minimum cuts ⟷ flow and cut-polyhedral geometry. The graph-theoretic path sees a lattice of minimum cuts and canonical residual-network representatives. The convex path sees optimal faces and dual certificates of a cut-defined polyhedron. The same equality cases are being organized in two languages.

Exercise 4.6. Show that the family of minimum s–t cut sides is closed under intersection and union even for arbitrary nonnegative arc capacities.

Remark (A larger Menger project). The Gomory–Hu direction is one of several follow-up projects collected in Section 16. That project menu is designed so that an individual student or a group may choose one direction—for example multiflow, Gomory–Hu trees, planar duality, gammoids, or infinite Menger—and develop it substantially without having to cover the other directions.

5 Edmonds' disjoint arborescences theorem

A spanning *r-arborescence* is a spanning directed tree rooted at r in which every vertex is reachable from r. We use $\rho_H(X)$ for the number of arcs of a digraph H entering X.

Theorem 5.1 (Edmonds)**.** *Let $D = (V, A)$, let $r \in V$, and let k be a nonnegative integer. There exist k pairwise arc-disjoint spanning r-arborescences if and only if*

$$\rho_D(X) \geq k \qquad \text{for every nonempty } X \subseteq V \setminus \{r\}. \tag{21}$$

This is Edmonds' disjoint-branchings theorem in the common-root case; see Edmonds [14]. The tight-set proof below is the Lovász proof technique [40]; see also Schrijver [60, Ch. 53] and Frank [22].

UNCROSSING CASE STUDY 3/8 — Edmonds arborescences (complete proof). A partial arborescence is extended one arc at a time while preserving the residual cut condition needed for the remaining $k - 1$ arborescences. Residual tight in-cuts are the only possible obstructions. Choosing a minimal tight set and uncrossing it with any alleged obstruction produces a smaller tight set, so a safe extension arc must exist.

5.1 Complete tight-set proof

Proof. Necessity is immediate: every spanning r-arborescence has an arc entering each nonempty set X not containing r, and arc-disjoint arborescences contribute distinct entering arcs.

For sufficiency we induct on k. The case $k = 0$ is vacuous. Assume $k \geq 1$, assume (21), and suppose the theorem is known for $k-1$. It is enough to construct one spanning r-arborescence T such that, after deleting its arcs, the residual digraph still satisfies

$$\rho_{D-A(T)}(X) \geq k-1 \qquad \text{for every nonempty } X \subseteq V \setminus \{r\}. \tag{22}$$

Indeed, the induction hypothesis then supplies $k-1$ further arc-disjoint spanning r-arborescences in $D - A(T)$.

We build T gradually. Let F be a partial out-arborescence rooted at r, let W be its vertex set, and write $R = D - A(F)$. Maintain the invariant $\rho_R(X) \geq k-1$ for every nonempty $X \subseteq V \setminus \{r\}$. It holds when F consists only of r, because (21) is stronger.

Call such a set X *tight* (with respect to R) when $\rho_R(X) = k-1$. Observe first that every tight set meets W. Indeed, since $\rho_D(X) \geq k$ whereas $\rho_R(X) = k-1$, at least one arc of F enters X; both ends of every arc of F lie in W.

Suppose $W \neq V$. If no tight set has a vertex outside W, choose any arc uv with $u \in W$ and $v \notin W$. Such an arc exists because the nonempty set $V \setminus W$ does not contain r, so $\rho_D(V \setminus W) \geq k > 0$. This arc enters no tight set, because its head lies outside W.

Otherwise choose, by inclusion, a minimal tight set T not contained in W. By the observation above, both $T \cap W$ and $T \setminus W$ are nonempty. We claim that there is an arc uv with $u \in T \cap W$ and $v \in T \setminus W$. If not, every arc of R entering $T \setminus W$ would already enter T, and hence $\rho_R(T \setminus W) \leq \rho_R(T) = k-1$. No arc of F enters $T \setminus W$, since all heads of F lie in W. Thus $\rho_D(T \setminus W) = \rho_R(T \setminus W) \leq k-1$, contradicting (21).

Fix such an arc uv. We next show that it enters no tight set. Suppose, to the contrary, that uv enters a tight set X. Then $v \in X \cap T$ while $u \in T \setminus X$. In particular, $X \cap T$ is nonempty, is a proper subset of T, and is not contained in W. Since the in-cut function of R is submodular,

$$2(k-1) = \rho_R(X) + \rho_R(T) \geq \rho_R(X \cap T) + \rho_R(X \cup T).$$

Both $X \cap T$ and $X \cup T$ are nonempty and avoid r, so the invariant gives a lower bound $k-1$ for each term on the right. Equality must therefore hold throughout, and $X \cap T$ is tight. This contradicts the minimal choice of T.

Now add uv to F. The new arc attaches a new vertex v, so the result is again a partial out-arborescence. If a set X is not entered by uv, its residual in-cut does not change. If uv enters X, then X was not tight; since in-cut sizes are integral, $\rho_R(X) \geq k$, and deleting uv leaves at least $k-1$ entering arcs. Thus the invariant is preserved.

Each extension increases $|W|$ by one, so after finitely many steps F becomes a spanning r-arborescence T satisfying (22). The induction hypothesis applied to $D - A(T)$ completes the proof. □

5.2 The arborescence polytope

For $k = 1$, consider

$$P_r(D) = \left\{ x \in \mathbb{R}^A_{\geq 0} : x(\delta^-(X)) \geq 1 \ \forall \varnothing \neq X \subseteq V \setminus \{r\}, \quad x(A) = |V| - 1 \right\}.$$

Edmonds' arborescence-polytope theorem (see Edmonds [14]) states that this polytope is integral: whenever it is nonempty, its vertices are exactly the incidence vectors of spanning r-arborescences. The preceding existence theorem identifies when the polytope is nonempty; the stronger integrality assertion is a separate theorem.

The exponentially many inequalities $x(\delta^-(X)) \geq 1$ look forbidding. At an extreme point, however, uncrossing lets us choose a laminar subfamily of tight inequalities that determines

the point. A laminar family has only $O(|V|)$ members. This is a typical reason why a polyhedron with exponentially many constraints can still be combinatorially rigid.

BRIDGE: Edmonds' tight-set proof $\longleftrightarrow$ the arborescence polytope. On the graph side, tight in-cuts control which arc can safely be added. On the polyhedral side, a laminar family of the same tight in-cut inequalities controls an extreme point. "Safe extension" and "integral vertex" are two manifestations of the same sparse tight-set geometry.

Exercise 5.2. Show that any laminar family of nonempty proper subsets of an n-element set has at most $2n - 2$ members.

6 Frank's free-root branching theorem

A closely related directed packing theorem allows the roots of the spanning arborescences to vary. The result is due to Frank [19]; see also Frank [22].

UNCROSSING CASE STUDY 4/8 — Frank free-root branchings (complete proof). A new root is joined to the original digraph by auxiliary arcs, and these arcs are deleted until the rooted Edmonds cut condition is inclusion-minimal. Every surviving auxiliary arc then has a tight-set certificate. Intersecting certificates are merged by uncrossing until they form a subpartition; the subpartition hypothesis forces exactly k auxiliary arcs to survive.

Theorem 6.1 (Frank)**.** *Let $D = (V, A)$ be a finite digraph and let $k \in \mathbb{Z}_{\geq 0}$. Then D contains k pairwise arc-disjoint spanning arborescences, whose roots may be different, if and only if every subpartition $\mathcal{F} = \{X_1, \dots, X_t\}$ of V satisfies*

$$\sum_{i=1}^{t} \rho_D(X_i) \geq k(t-1). \tag{23}$$

Here a *subpartition* is a family of pairwise disjoint nonempty subsets; it need not cover V.

Proof. For necessity, let $B_1, \dots, B_k$ be pairwise arc-disjoint spanning arborescences, with arbitrary roots. Fix a subpartition $\mathcal{F} = \{X_1, \dots, X_t\}$. In one arborescence B_j, at most one member of $\mathcal{F}$ contains its root. Every other member must therefore receive an entering arc of B_j. Because the members of $\mathcal{F}$ are disjoint, these entering arcs are distinct, so B_j contributes at least $t - 1$ arcs to the sum of the in-cuts. Summing over the k arc-disjoint arborescences gives (23).

For sufficiency, adjoin a new vertex r. Initially add k parallel arcs rv for every $v \in V$; call these *root arcs*. The resulting digraph certainly satisfies

$$\rho(X) \geq k \qquad \text{for every nonempty } X \subseteq V. \tag{24}$$

Delete root arcs one at a time as long as (24) remains true, and let D' be a resulting inclusion-minimal augmentation. Let R denote its surviving root arcs and put $m = |R|$. Taking $X = V$ in (24) gives $m = \rho_{D'}(V) \geq k$.

We prove the reverse inequality. For each surviving root arc $a = rv \in R$, minimality says that deleting a would violate (24). Hence there is a nonempty set $X_a \subseteq V$ with $v \in X_a$ and $\rho_{D'}(X_a) = k$. Call these sets tight.

If two tight sets X and Y intersect, then their union is tight. Indeed, submodularity of the in-cut function and (24) give

$$2k = \rho_{D'}(X) + \rho_{D'}(Y) \geq \rho_{D'}(X \cap Y) + \rho_{D'}(X \cup Y) \geq 2k.$$

Thus equality holds throughout. Starting with one certificate X_a for each $a \in R$, repeatedly replace two intersecting members by their union. The process terminates with pairwise disjoint tight sets $T_1, \dots, T_q$. Their union still contains the head of every surviving root arc.

Let s_i be the number of surviving root arcs whose heads lie in T_i. Since the T_i are disjoint and every surviving root-arc head lies in their union, $m = \sum_i s_i$. Moreover, the only arcs of D' that are not arcs of D are the root arcs, so tightness gives $k = \rho_{D'}(T_i) = \rho_D(T_i) + s_i$. Summing over i and using the subpartition hypothesis for $\{T_1, \dots, T_q\}$ yields

$$kq = \sum_{i=1}^{q} \rho_D(T_i) + m \geq k(q-1) + m.$$

Hence $m \leq k$. Together with $m \geq k$, we obtain $m = k$.

Now apply Edmonds' theorem, Theorem 5.1, to D' with root r. Condition (24) gives k pairwise arc-disjoint spanning r-arborescences. Each such arborescence must contain at least one arc leaving r. Since there are exactly k surviving root arcs altogether and the arborescences are arc-disjoint, each arborescence contains exactly one of them. Delete r and that unique root arc from each arborescence. What remains is a spanning arborescence of D, rooted at the head of the deleted root arc. The k resulting arborescences are still pairwise arc-disjoint, as required. □

7 Nash–Williams–Tutte tree packing

For a partition $\mathcal{P} = \{V_1, \dots, V_t\}$ of V, let $e_G(\mathcal{P})$ be the number of edges with endpoints in different parts.

Theorem 7.1 (Nash–Williams–Tutte)**.** *Let $G = (V, E)$ be a finite loopless undirected multigraph and let $k \in \mathbb{Z}_{\geq 0}$. Then G contains k pairwise edge-disjoint spanning trees if and only if*

$$e_G(\mathcal{P}) \geq k(t-1)$$

for every partition $\mathcal{P}$ of V into t nonempty parts.

The theorem was proved independently by Nash–Williams [48] and Tutte [62]. There are several routes to it. We first give a complete proof through rooted orientations and Edmonds' arborescence theorem. This route is especially well suited to the present notes because its key step is another tight-set uncrossing argument. Afterwards we return to the matroid-union proof, which makes the partition condition especially transparent from the matroid side.

UNCROSSING CASE STUDY 5/8 — rooted orientations and the Nash–Williams–Tutte theorem (complete proof). Hall's theorem first reappears as an orientation lemma prescribing indegrees. A minimal feasible indegree vector has tight vertex sets; intersecting tight sets uncross, and their maximal members form a partition. The partition inequality then forces the total indegree to be exactly $|E|$, producing a rooted orientation with at least k entering arcs at every set avoiding the root. Edmonds' theorem converts that orientation into k edge-disjoint spanning trees.

For $X \subseteq V$, let $i_G(X)$ denote the number of edges with both ends in X, and for $m : V \to \mathbb{Z}_{\geq 0}$ write $m(X) = \sum_{v \in X} m(v)$.

Lemma 7.2 (Orientation lemma)**.** *Let $G = (V, E)$ be a loopless multigraph and let $m : V \to \mathbb{Z}_{\geq 0}$. There is an orientation D of G with $d^-_D(v) = m(v)$ for every v if and only if $m(V) = |E|$ and $m(X) \geq i_G(X)$ for every $X \subseteq V$.*

Proof. Necessity is immediate: the sum of all indegrees is $|E|$, while the edges induced by X already contribute $i_G(X)$ units to the indegrees of vertices in X.

For sufficiency, create a bipartite graph whose left side is E and whose right side contains $m(v)$ copies of each vertex v. An edge $e = uv$ on the left is adjacent to all copies of u and v. If $F \subseteq E$ and W is the set of endpoints of edges in F, then $F \subseteq E(G[W])$, so $|F| \le i_G(W) \le m(W) = |N(F)|$. Hall's theorem therefore gives a matching saturating the left side. Because the two sides have the same total size, every right-side copy is also matched. Orient each original edge toward the endpoint-copy to which it is matched. Exactly $m(v)$ edges are then oriented into v. □

Theorem 7.3 (Frank's rooted orientation theorem [18])**.** *Let $G = (V, E)$ be a finite loopless undirected multigraph, let $r \in V$, and let $k \in \mathbb{Z}_{\ge 0}$. Then G has an orientation D satisfying $\rho_D(X) \ge k$ for every nonempty $X \subseteq V \setminus \{r\}$ (we call this a rooted k-in-connected orientation at r) if and only if $e_G(\mathcal{P}) \ge k(|\mathcal{P}| - 1)$ for every partition $\mathcal{P}$ of V.*

Proof. Necessity is immediate. If $\mathcal{P}$ has q parts, then every part not containing r must receive at least k entering arcs. Summing their indegrees counts only cross-edges and gives $e_G(\mathcal{P}) \ge k(q-1)$.

For sufficiency, loops may be deleted. Define

$$p(X) = \begin{cases} i_G(X) + k, & \varnothing \ne X \subseteq V \setminus \{r\}, \\ i_G(X), & r \in X, \\ 0, & X = \varnothing. \end{cases}$$

The function i_G is supermodular, and the extra k is preserved on both sides of the supermodular inequality whenever two nonempty sets intersect. Hence p is intersecting supermodular.

Choose an integer vector $m : V \to \mathbb{Z}_{\ge 0}$ with $m(X) \ge p(X)$ for every $X \subseteq V$, and among all such vectors choose one minimizing $m(V)$. Such vectors certainly exist if the coordinates are chosen sufficiently large. Call X *tight* when $m(X) = p(X)$. Every vertex belongs to a tight set. Indeed, if a non-root vertex v belonged to none, then $m(v) > 0$ and decreasing $m(v)$ by 1 would preserve all inequalities, contradicting minimality. For the root, either $m(r) > 0$ and the same argument applies, or $m(r) = 0 = p(\{r\})$, so $\{r\}$ is tight.

If tight sets X and Y intersect, then $X \cap Y$ and $X \cup Y$ are tight. Since m is modular and p is intersecting supermodular,

$$\begin{aligned} m(X \cap Y) + m(X \cup Y) &= m(X) + m(Y) = p(X) + p(Y) \\ &\le p(X \cap Y) + p(X \cup Y) \\ &\le m(X \cap Y) + m(X \cup Y). \end{aligned}$$

Thus equality holds throughout. It follows that the inclusion-maximal tight sets are pairwise disjoint; since every vertex lies in a tight set, they form a partition $\mathcal{P} = \{V_1, \dots, V_q\}$ of V.

Now

$$m(V) = \sum_{j=1}^{q} p(V_j) = \sum_{j=1}^{q} i_G(V_j) + k(q-1) = |E| - e_G(\mathcal{P}) + k(q-1) \le |E|.$$

On the other hand, $m(V) \ge p(V) = i_G(V) = |E|$, so $m(V) = |E|$. Since $m(X) \ge p(X) \ge i_G(X)$ for every X, the orientation lemma gives an orientation D with indegree vector m. For every nonempty $X \subseteq V \setminus \{r\}$, $m(X) = i_G(X) + \rho_D(X)$, and therefore $\rho_D(X) = m(X) - i_G(X) \ge p(X) - i_G(X) = k$. This is the desired rooted orientation. □

Complete proof of Theorem 7.1. Necessity is immediate: after contracting the parts of a partition into q vertices, every spanning tree still has to connect those q vertices and therefore uses at least $q - 1$ cross-edges.

For sufficiency, fix an arbitrary root $r \in V$. By Theorem 7.3, the partition inequalities orient G so that every nonempty $X \subseteq V \setminus \{r\}$ has at least k entering arcs. Edmonds' theorem, Theorem 5.1, then gives k pairwise arc-disjoint spanning r-arborescences. Forgetting the directions produces k pairwise edge-disjoint spanning trees of G. □

7.1 A second proof through matroid union

The preceding argument proves the tree-packing theorem using cut inequalities, orientations, and arborescences, so it belongs to the *general submodular cut* side of the hierarchy. The following second proof explains the same partition condition from the *matroid* side through the graphic matroid and matroid union. This is a useful example of one theorem having two distinct structural homes.

Theorem 7.4 (Edmonds' matroid union theorem, rank formula)**.** *Let $M = (E, \mathcal{I})$ be a matroid with rank function r. A set is independent in the union of k copies of M if it can be partitioned into k independent sets of M. The rank of this union matroid is*

$$r_k(E) = \min_{F \subseteq E} \bigl(|E \setminus F| + k\, r(F)\bigr).$$

We shall use the theorem as a black box; it is one of the foundational min–max theorems of matroid theory. See, for example, Schrijver [60].

7.2 Graphic matroids and the spanning-tree polytope

The graphic matroid $M(G)$ has ground set E and independent sets equal to forests. Its bases are spanning trees when G is connected. The spanning-tree polytope is

$$\mathrm{ST}(G) = \mathrm{conv}\{\mathbf{1}_T : T \text{ spanning tree}\}.$$

Edmonds' description is

$$\mathrm{ST}(G) = \Bigl\{x \in \mathbb{R}^E_{\geq 0} : x(E) = |V| - 1, \quad x(E(S)) \leq |S| - 1 \ \forall \varnothing \neq S \subsetneq V\Bigr\}.$$

The theorem says this polytope is integral.

Packing k edge-disjoint spanning trees asks for k points of this polytope whose sum is componentwise at most 1. The partition inequalities in Theorem 7.1 are exactly the obstructions that survive after uncrossing the corresponding rank inequalities. Matroid union makes this precise.

BRIDGE: spanning-tree packing $\longleftrightarrow$ graphic-matroid and zonotope geometry. The packing theorem is expressed by partition inequalities and matroid union; earlier, the same graphic matroid appears through its base polytope, while spanning trees index determinant/volume contributions of the graphical zonotope. Thus one combinatorial object simultaneously controls a min–max packing theorem and a convex-geometric decomposition. The first body is matroidal; the second comes from the generally nonmonotone graph cut function and therefore lies in the larger submodular world.

Matroid-union proof of Theorem 7.1. Necessity is immediate: each spanning tree uses at least $t - 1$ edges between the t parts of any partition.

For sufficiency, let $M(G)$ be the graphic matroid and assume the partition inequalities. Fix $F \subseteq E$, and let $\mathcal{P}_F$ be the partition of V into the connected components of (V, F). If $t = |\mathcal{P}_F|$, then $r_{M(G)}(F) = |V| - t$. Every edge joining two different components of (V, F) lies in $E \setminus F$, so by the Nash–Williams–Tutte hypothesis

$$|E \setminus F| \geq e_G(\mathcal{P}_F) \geq k(t - 1).$$

Consequently,

$$|E \setminus F| + k r_{M(G)}(F) \geq k(t - 1) + k(|V| - t) = k(|V| - 1).$$

The matroid union theorem therefore gives a set of $k(|V| - 1)$ edges that can be partitioned into k forests. Each forest has at most $|V| - 1$ edges, so all k forests have exactly $|V| - 1$ edges. Hence each is a spanning tree, and they are edge-disjoint. □

For further proofs, compare the classical forest-exchange proof in Diestel [11, Sec. 2.4] and Kaiser's short elementary proof [38].

Partition conditions are compressed cut conditions. A single cut inequality only checks two parts. Packing a spanning object forces compatibility across many parts simultaneously. The Nash–Williams–Tutte condition is what remains when all such requirements are aggregated into a partition. Submodularity is the mechanism that says one need not inspect arbitrary overlapping collections independently.

Corollary 7.5. *Every $2k$-edge-connected graph contains k edge-disjoint spanning trees.*

Proof. For a partition into t parts, every part has cut size at least $2k$. Summing these cut sizes counts each cross-edge twice, hence $2e_G(\mathcal{P}) \geq 2kt$, so $e_G(\mathcal{P}) \geq kt \geq k(t-1)$. □

Exercise 7.6 (Arboricity)**.** The Nash–Williams arboricity theorem states that the minimum number of forests whose union is $E(G)$ equals

$$\max_{H \subseteq G,\ |V(H)| \geq 2} \left\lceil \frac{|E(H)|}{|V(H)| - 1} \right\rceil .$$

Relate this to matroid union and compare its density condition with Theorem 7.1.

Part III
Orientations, splitting-off, and local transformations

8 Robbins and Nash–Williams: orienting connectivity

Theorem 8.1 (Robbins, 1939)**.** *Every finite loopless undirected multigraph has a strongly connected orientation if and only if it is 2-edge-connected.*

Proof. If an orientation of G is strongly connected, then G has no bridge. Indeed, if a bridge e separates V into two nonempty parts S and $V \setminus S$, then whichever direction is assigned to e, no directed path can cross the cut in the opposite direction. Thus G must be 2-edge-connected.

Conversely, assume that G is 2-edge-connected. Run depth-first search from a root r and let T be the resulting DFS spanning tree. Orient every tree edge away from r. In an undirected DFS, every non-tree edge joins a vertex to one of its ancestors in T; orient each such edge from the descendant toward the ancestor.

The root r reaches every vertex along the directed tree edges. It remains to show that every vertex reaches r. For a vertex $v \neq r$, let S_v be the vertex set of the subtree of T rooted at v, and let $p(v)$ be the parent of v. The tree edge $p(v)v$ is not a bridge, so the cut $\delta(S_v)$ contains another edge xy. This second edge cannot be a tree edge, because $p(v)v$ is the unique tree edge joining S_v to its complement. Hence xy is a non-tree edge. By the DFS property, one endpoint is an ancestor of the other. Since one endpoint lies in S_v and the other lies outside, we may write $x \in S_v$ and $y \notin S_v$, where y is a strict ancestor of v; by construction the edge is oriented $x \to y$.

Starting at v, follow tree edges downward to x, then take the arc $x \to y$. We have therefore reached a vertex y of strictly smaller depth than v. Repeating this argument strictly decreases the depth, so eventually we reach r. Thus every vertex can reach r, while r can reach every vertex, and the orientation is strongly connected. □

The proof is constructive and yields a linear-time orientation once a DFS tree is known. See Robbins' original paper [54]; standard graph-theory texts such as Bondy–Murty [6] also discuss this orientation theorem.

A deeper generalization is the following.

Theorem 8.2 (Nash–Williams orientation theorem)**.** *Let G be a finite loopless undirected multigraph and let $k \in \mathbb{Z}_{\geq 0}$. Then G admits a k-arc-strong orientation if and only if G is $2k$-edge-connected.*

Necessity is immediate: if a cut has a arcs entering and b arcs leaving, a k-arc-strong orientation requires $a, b \geq k$, so the underlying cut has size at least $2k$.

We shall see three proof philosophies: splitting-off, convex integrality, and local edge flips. They are kept in consecutive sections because they prove the *same* theorem from three different structural viewpoints; this is a deliberate comparison block rather than three unrelated applications.

9 Splitting-off and constructive characterizations

9.1 A minimal connectivity lemma

UNCROSSING CASE STUDY 6/8 — Mader's degree lemma (complete proof). This is the cleanest "minimal counterstructure" version of uncrossing in the handout: two crossing tight cuts would produce a strictly smaller tight cut, contradicting minimality. Both submodularity and posimodularity are needed to handle the two possible geometries of the crossing pair.

Theorem 9.1 (Mader's degree lemma)**.** *For $k \geq 1$, every edge-minimal finite loopless k-edge-connected undirected multigraph has a vertex of degree exactly k.*

Proof. Every edge belongs to a cut of size exactly k, for otherwise that edge could be deleted. Choose a nonempty proper $X \subset V$ with $d(X) = k$ and $|X|$ minimum.

Suppose $|X| > 1$. Then $G[X]$ is connected; otherwise each component of $G[X]$ would have cut size at least k, forcing $d(X) \geq 2k > k$. Choose an edge $e = uv$ inside X. Since $G - e$ is not k-edge-connected, there is a set Y whose cut drops below k after deleting e. Hence $d_G(Y) = k$ and exactly one of u, v lies in Y. Thus $X \cap Y \neq \varnothing$ and $X \setminus Y \neq \varnothing$.

If $Y \subsetneq X$, then Y is already a smaller nonempty k-tight set, a contradiction. Otherwise $Y \setminus X \neq \varnothing$, so X and Y overlap. If $X \cup Y \neq V$, submodularity gives

$$2k = d(X) + d(Y) \geq d(X \cap Y) + d(X \cup Y) \geq 2k,$$

so $d(X \cap Y) = k$, contradicting the minimality of X. If $X \cup Y = V$, posimodularity gives

$$2k \geq d(X \setminus Y) + d(Y \setminus X) \geq 2k,$$

so $d(X \setminus Y) = k$, again a smaller nonempty subset of X. □

This proof is a model example of uncrossing: minimality plus a tight cut forces crossing to disappear.

9.2 Splitting-off

Let z be a vertex and let uz, zv be two incident edges. *Splitting off* this pair means deleting uz, zv and adding uv (possibly creating a parallel edge). A *complete splitting-off* pairs all edges at an even-degree vertex and splits every pair, after which the isolated vertex is deleted.

UNCROSSING CASE STUDY 7/8 — Lovász splitting-off (complete proof). An inadmissible pair is certified by a nearly tight dangerous set. Uncrossing dangerous sets lets one choose a maximal certificate; parity and the degree condition then force a neighbor outside it, yielding a safe pair. This is the most literal later incarnation of Hall's "decompose or make a safe local move" principle.

Theorem 9.2 (Lovász splitting-off, connectivity form)**.** *Let $k \geq 1$, let G be a finite loopless multigraph, and let z be a vertex of degree $2k$. Assume that*

$$d_G(X) \geq 2k \qquad (\varnothing \neq X \subsetneq V(G) \setminus \{z\}).$$

In particular, this hypothesis holds when G itself is $2k$-edge-connected. Then the edges incident with z can be paired so that complete splitting-off at z leaves a $2k$-edge-connected loopless multigraph on $V(G) \setminus \{z\}$ (loops created during splitting may be deleted).

A stronger local-connectivity theorem is known, but this form is enough for Nash–Williams.

9.3 Dangerous sets and uncrossing

Fix one edge zv. A second edge zu is *inadmissible* if splitting zv, zu destroys $2k$-edge-connectivity. For $X \subseteq V \setminus \{z\}$, write $d(X)$ for its undirected cut size. Directly inspecting the split shows that inadmissibility is equivalent to the existence of a proper set $X \subsetneq V \setminus \{z\}$ with $u, v \in X$ and $d(X) \leq 2k+1$; call such an X *dangerous* (for the fixed edge zv).

We first prove the uncrossing fact needed below. If dangerous sets X, Y containing v are nested, their union is dangerous. Otherwise put $A = X \setminus Y$, $B = Y \setminus X$, $C = X \cap Y$, and $D = V \setminus (X \cup Y)$. The cut-counting identity

$$d(X) + d(Y) = d(A) + d(B) + 2|E(C, D)|$$

holds. Since A, B are nonempty proper sets, $d(A), d(B) \geq 2k$; since zv joins C to D, $|E(C, D)| \geq 1$. Thus equality is forced throughout: $d(A) = d(B) = 2k$, $|E(C, D)| = 1$, and $d(X) + d(Y) = 4k + 2$. Hence $d(X) = d(Y) = 2k + 1$. Submodularity, in its exact cut-counting form, gives $d(X \cup Y) \leq 2k + 2$. On the other hand, the three cuts of X, $B = Y \setminus X$, and $X \cup Y$ satisfy

$$d(X \cup Y) \equiv d(X) + d(B) \equiv 1 \pmod 2;$$

this follows by counting every edge incident with the three regions modulo two. Since the degree of z remains even throughout splitting-off, $X \cup Y$ cannot equal $V \setminus \{z\}$. The invariant therefore gives $d(X \cup Y) \geq 2k$, and parity forces $d(X \cup Y) = 2k + 1$. Consequently the union M of all dangerous sets containing v is itself a proper dangerous set.

If every neighbor of z belonged to M, put $W = V \setminus (M \cup \{z\})$, which is nonempty because M is proper. There are no edges from z to W, so the invariant gives $|E(M, W)| = d(W) \geq 2k$. All remaining edges at z go to M, and at least the fixed edge zv and one prospective partner remain; hence $d(z) \geq 2$. It follows that

$$d(M) = |E(M, W)| + d(z) \geq 2k + 2,$$

contrary to the fact that M is dangerous. Choose a neighbor $u \notin M$. By construction no dangerous set contains both u and v, so the pair zv, zu is admissible. Split it, delete the resulting loop if one is created, and repeat. At every stage the invariant $d(X) \geq 2k$ for all nonempty $X \subsetneq V \setminus \{z\}$ is preserved, while the degree of z drops by two. After k admissible splits, z is isolated and can be deleted.

Classroom derivation. Draw the four regions $X \cap Y$, $X \setminus Y$, $Y \setminus X$, and $V \setminus (X \cup Y)$. Assign a variable to the number of edges between each pair of regions. Derive both submodularity and posimodularity by cancellation. Then identify exactly which inter-region edges measure the slack in the inequalities. This turns "uncrossing" from a slogan into a visible counting identity.

9.4 Pinching: the inverse operation

To *pinch k edges $u_i v_i$* with a new vertex z, subdivide each of the k chosen edges and identify the k subdivision vertices. Thus each $u_i v_i$ is replaced by $u_i z$ and $z v_i$, and the new vertex has degree $2k$.

Theorem 9.3 (Lovász constructive characterization)**.** *For $k \geq 1$, a finite loopless undirected multigraph is $2k$-edge-connected if and only if it can be obtained from a single vertex by repeatedly*

1. *adding an edge between existing vertices, or*
2. *pinching k existing edges with one new vertex.*

Derivation. Delete edges until the graph is edge-minimal while preserving $2k$-edge-connectivity. By Theorem 9.1, there is a vertex z of degree $2k$. By Theorem 9.2, split off z completely, producing a smaller $2k$-edge-connected graph. Induct and reverse the operations. □

9.5 The splitting-off proof of Nash–Williams

First proof of Theorem 8.2. Induct along the construction in Theorem 9.3. The one-vertex graph is trivial. Adding an edge cannot decrease any directed cut value, so it may be oriented arbitrarily.

Suppose that the construction pinches already oriented edges $u_i \to v_i$ and replaces each by the directed path $u_i \to z \to v_i$. Let D be the old k-arc-strong orientation and D' the new one. If a nonempty proper set X does not contain z, every old arc entering (respectively leaving) X is replaced by a path contributing at least one arc entering (respectively leaving) X, regardless of the side on which z lies. Thus both directed boundary sizes of X in D' are at least those in D.

If $z \in X$, put $Y = V(D) \setminus X$. The arcs leaving Y in D correspond to paths contributing an entering arc of X in D', and the arcs entering Y correspond to paths contributing a leaving arc of X. Since D is k-arc-strong, both quantities are at least k. Hence every nontrivial cut of D' has at least k entering and at least k leaving arcs. The pinching operation therefore preserves k-arc-strong connectivity, and induction proves the theorem. □

10 The convex-integrality proof of Nash–Williams

This proof makes the "submodularity as convexity" theme completely explicit.

Start from an arbitrary orientation $D = (V, A)$ of the undirected $2k$-edge-connected graph G. For each arc $a \in A$, introduce a variable $x_a \in [0, 1]$, interpreted as whether we reverse a. For a set $X \subset V$, after reversals the number of arcs entering X is $\rho_D(X) - x(\delta_D^-(X)) + x(\delta_D^+(X))$. We want this to be at least k, so

$$x(\delta_D^-(X)) - x(\delta_D^+(X)) \leq \rho_D(X) - k \qquad (\varnothing \neq X \subsetneq V).$$

The integral submodular-flow theorem of Edmonds–Giles [15] (in the cut-system form used in Frank's monograph [22]) applies to this system: a crossing-submodular cut bound, together with integral lower and upper bounds on the arc variables, defines an integral polyhedron. Here $b(X) = \rho_D(X) - k$ is crossing submodular because ρ_D is submodular, and the bounds

$0 \le x_a \le 1$ are integral. We use the theorem only for this finite network instance, with the usual boundary-set conventions for the cut function.

Now take the fractional point with $x_a = 1/2$ for every $a \in A$. For a nontrivial cut X, let $h := \rho_D(X)$ and $\ell := |\delta_D^+(X)|$. At $x_a = 1/2$, the left-hand side of (10) is $(h - \ell)/2$, so the inequality is equivalent to $h + \ell \ge 2k$. The quantity $h + \ell$ is exactly the size of the underlying undirected cut, which is at least $2k$.

Therefore the polyhedron is nonempty. Integrality yields a feasible 0–1 point x. Reverse exactly those arcs with $x_a = 1$. The resulting orientation has at least k arcs entering every nontrivial set. Applying the same inequality to the complement gives at least k leaving arcs as well, so the orientation is k-arc-strong (equivalently, it has k arc-disjoint directed paths between every ordered pair by Menger's theorem).

This proof has an unusually transparent convex message:

> The symmetric fractional orientation "half each way" is obviously feasible. Submodular-flow integrality says that feasibility of this convex relaxation forces feasibility of an integral orientation.

The graph theorem is therefore an integrality theorem in disguise.

Exercise 10.1. Rewrite the system (10) in terms of a signed incidence matrix. Compare it with an ordinary circulation polyhedron. Which part is new, and why is submodularity the correct replacement for ordinary node-balance constraints?

11 A modern local proof: monotone edge flips

A striking refinement asks whether one can reach a highly connected orientation by changing one edge at a time. The result below is due to Ito, Iwamasa, Kakimura, Kamiyama, Kobayashi, Maezawa, Nozaki, Okamoto, and Ozeki [35].

Theorem 11.1 (Ito et al. [35])**.** *For $k \ge 2$, from every orientation of a finite loopless undirected graph that is $(2k + 2)$-edge-connected, one can flip edges one by one so that directed edge-connectivity never decreases, and eventually obtain a k-arc-strong orientation. The same paper proves that the graph of k-arc-strong orientations is connected under the $(2k+2)$-edge-connectivity hypothesis.*

This gives a dynamical proof of Nash–Williams: instead of jumping from a fractional point to an integral point, one walks through the discrete space of orientations by local moves.

There are now three complementary geometries of the same theorem:

1. **splitting geometry:** reduce the graph while preserving cuts;
2. **polyhedral geometry:** move from the half-vector to an integral feasible point;
3. **reconfiguration geometry:** move through the graph of orientations by edge flips.

A useful research habit is to ask whether a theorem known in one geometry has a stronger analogue in the other two.

BRIDGE: Nash–Williams orientation $\longleftrightarrow$ splitting geometry and convex integrality. The same orientation theorem has already been proved by splitting-off and by an integral submodular-flow polyhedron; the monotone edge-flip theorem of Ito et al. [35] adds a third, reconfiguration-theoretic geometry. This is the clearest place in the handout where

the two principal reading paths prove the same theorem by genuinely different mechanisms.

12 Mader's directed theorem

For a digraph, pinching k arcs means subdividing k arcs and identifying the subdivision vertices, so the new vertex has indegree k and outdegree k.

Theorem 12.1 (Mader). *Let $k \geq 1$. A finite loopless digraph is k-arc-strong if and only if it can be obtained from a single vertex by repeatedly*

1. *adding an arc, or*
2. *pinching k existing arcs with one new vertex.*

The proof rests on two structural results.

Theorem 12.2 (Directed splitting-off). *Let $k \geq 1$ and let $D = (V \cup \{z\}, A)$ be a finite loopless digraph such that every nontrivial cut of D has at least k entering and k leaving arcs. Suppose that z is balanced, meaning that its indegree equals its outdegree. Then one can pair arcs entering z with arcs leaving z so that complete directed splitting-off preserves k-arc-strong connectivity on V.*

UNCROSSING CASE STUDY 8/8 — Mader's balanced-vertex theorem (proof outline). This is the most sophisticated laminar argument in the handout. A minimal cover by tight in-cuts is uncrossed into nested structure; after separating sets according to a fixed vertex, minimal members force strong subdigraphs, and a second minimality argument finally produces a balanced singleton.

Theorem 12.3 (Mader's balanced-vertex theorem). *For $k \geq 1$, every arc-minimal finite loopless k-arc-strong digraph has at least two vertices v with $d^-(v) = d^+(v) = k$.*

The proof of Theorem 12.3 is a particularly sophisticated application of uncrossing. One takes a minimal family of tight in-cuts that covers every arc, uncrosses it until intersecting members are nested, then fixes a vertex s and separates the family into sets not containing s and complements of sets containing s. Minimal members induce strong subdigraphs. A second minimality argument forces a singleton that is simultaneously tight for indegree and outdegree, yielding a balanced vertex.

Suggested seminar exercise. Reconstruct the proof of Theorem 12.3 from the following four invariants:

1. every arc enters some tight set;
2. crossing tight in-cuts may be uncrossed;
3. after choosing s, every arc either enters a member of one laminar family or leaves a member of another;
4. a minimal nonsingleton member induces a strongly connected subdigraph.

The final singleton argument is an excellent training problem in how global laminar structure forces a local degree statement.

Part IV
General frameworks and further connections

13 Submodular flows, Lucchesi–Younger, and Woodall's conjecture

The convex proof of Nash–Williams sits inside the much broader theory of submodular flows, developed by Edmonds, Giles, Frank, Fujishige, and others. In the hierarchy introduced after Theorem 1.5, this theory lies *beyond polymatroids*: monotonicity of the governing set function is not the essential feature, and crossing-submodular systems naturally arise. Polymatroid intersection is an important special ancestor, while submodular flow incorporates network structure together with more general submodular constraints.

A typical submodular-flow system replaces ordinary conservation equalities by set-indexed inequalities of the form $x(\delta^-(X)) - x(\delta^+(X)) \le b(X)$, where b is submodular. Under suitable crossing-submodularity hypotheses, the feasible polyhedron is integral when the data are integral.

For a digraph $D = (V, A)$ and $X \subseteq V$, write $\delta^+(X)$ and $\delta^-(X)$ for the arcs leaving and entering X, respectively. A *dicut* is a set $\delta^+(X)$ with $\varnothing \neq X \subsetneq V$ and $\delta^-(X) = \varnothing$; it is *nonempty* when $\delta^+(X) \neq \varnothing$. Thus every arc crossing the corresponding vertex cut points from X to $V \setminus X$. Some references use the opposite orientation convention; replacing X by $V \setminus X$ makes the two conventions equivalent. A *dijoin* is a set of arcs meeting every nonempty dicut. With these conventions, the celebrated theorem of Lucchesi and Younger reads as follows.

Theorem 13.1 (Lucchesi–Younger [44])**.** *In every finite digraph, the minimum cardinality of a dijoin equals the maximum number of pairwise arc-disjoint nonempty dicuts.*

This is a directed packing–covering theorem at a more global level than the usual one-pair max-flow/min-cut theorem: the objects being packed are dicuts, whereas the covering objects are dijoins. Uncrossing again organizes optimal cut families.

There is a natural statement obtained by interchanging the two combinatorial roles. This role reversal should not be confused with linear-programming duality: it asks for a new integral packing theorem rather than merely restating the LP dual of Lucchesi–Younger. Assume that D has at least one nonempty dicut. Let $\tau(D)$ be the minimum size of such a dicut, and let $\nu(D)$ be the maximum number of pairwise arc-disjoint dijoins. Every dijoin must use an arc of every minimum dicut, so the elementary inequality $\nu(D) \le \tau(D)$ always holds.

Conjecture 13.2 (Woodall [64]). For every finite digraph with at least one nonempty dicut, $\nu(D) = \tau(D)$; equivalently, if every nonempty dicut has at least k arcs, then the digraph contains k pairwise arc-disjoint dijoins.

Status. As of September 2026, Woodall's conjecture remains open in general. The assertion is known when $\tau(D) \le 2$, but the case $\tau(D) = 3$ is still open even for planar digraphs [2]. The stronger capacitated formulation proposed by Edmonds and Giles is false: Schrijver constructed a 0–1 weighted digraph in which every dicut has total weight at least 2, but the arcs of weight 1 cannot be partitioned into two dijoins [59]. On the positive side, Abdi, Cornuéjols, and Zlatin proved that if $\tau = \tau(D)$, then the whole arc set can be partitioned into a dijoin and a $(\tau - 1)$-dijoin, where a q-dijoin meets every nonempty dicut in at least q arcs [1]. More recently, Cornuéjols, Liu, and Ravi proved that every digraph with minimum dicut size τ contains at least $\lfloor \tau/6 \rfloor$ pairwise arc-disjoint dijoins, and that such a packing can be found in polynomial time [9].

Thus Lucchesi–Younger and Woodall form a particularly instructive pair. The first min–max equality is a theorem with a strong submodular-flow explanation; the formally role-reversed equality is a long-standing open problem whose weighted analogue already fails. This contrast is

a useful warning that a beautiful integral duality need not remain true after simply swapping the packing and covering objects.

Project 13.3. Study Lucchesi–Younger and Woodall side by side. Give a precise account of the weak inequalities in both directions, identify where submodular-flow integrality proves Lucchesi–Younger, and explain why the same mechanism does not prove Woodall's conjecture. Then read one of the recent papers on dijoin packing and isolate a concrete partial theorem, special class, or approximation result that can be proved in detail.

14 Further beautiful mathematics around the same ideas

The examples in this section are not included merely as a list of concepts. Each subsection records at least one theorem showing what the relevant form of submodularity actually does. Some proofs are short enough to assign as exercises; the references point to broader treatments.

14.1 Coverage functions: from Hall to approximation algorithms

The Hall neighborhood function is a coverage function. More generally, let $\{A_u : u \in U\}$ be subsets of a weighted universe R, with weights $w_v \geq 0$, and define $f(X) := \sum_{v \in \bigcup_{u \in X} A_u} w_v$. Such a function is normalized, monotone, and submodular, hence it is a polymatroid rank function; with integer weights it is an integral polymatroid rank function. The unweighted Hall function $X \mapsto |N(X)|$ is the basic graph-theoretic example. In the bipartite-graph notation $A_u = N(u)$, its Lovász extension is $\widehat{f}(x) = \sum_{v \in R} w_v \max_{u \in N(v)} x_u$ for $x \geq 0$.

The importance of coverage functions goes far beyond Hall's theorem: they are the model example for monotone submodular maximization.

Theorem 14.1 (Nemhauser–Wolsey–Fisher greedy bound)**.** *Let $f : 2^U \to \mathbb{R}_{\geq 0}$ be monotone and submodular with $f(\varnothing) = 0$. Fix $k \geq 1$, and construct $S_0 = \varnothing$ and then $S_{i+1} = S_i \cup \{e_i\}$, where e_i has maximum marginal gain among elements outside S_i. If S^* maximizes $f(S)$ subject to $|S| \leq k$, then*

$$f(S_k) \geq \left(1 - \left(1 - \frac{1}{k}\right)^k\right) f(S^*) \geq (1 - e^{-1}) f(S^*).$$

The proof is a one-line use of diminishing returns repeated k times: at stage i, the best available marginal gain is at least $(f(S^*) - f(S_i))/k$. This theorem, including the sharper finite-k factor, is due to Nemhauser, Wolsey, and Fisher [49].

Exercise 14.2 (Greedy recurrence)**.** Prove Theorem 14.1 from the recurrence $f(S_{i+1}) - f(S_i) \geq (f(S^*) - f(S_i))/k$. Then specialize the argument to the maximum-coverage problem and identify exactly where coverage submodularity is used.

14.2 Entropy: submodularity and Shearer's inequality

For jointly distributed discrete random variables $(X_i)_{i \in U}$ with finite entropies, the entropy set function $h(S) := H(X_S)$ is normalized, monotone, and submodular. Thus it is a polymatroid rank function, often called an *entropic polymatroid*. Fujishige emphasized this polymatroidal dependence structure in probability and information theory [25]. Not every polymatroid is entropic, so this is a distinguished subclass rather than a characterization of all polymatroids. Equivalently, its submodularity slack is a conditional mutual information and is therefore nonnegative. One of the cleanest combinatorial consequences is Shearer's inequality.

Theorem 14.3 (Shearer's inequality)**.** *Let $X = (X_1, \dots, X_n)$ be a discrete random vector, and let $\mathcal{A}$ be a multiset of subsets of $[n]$ such that every coordinate i belongs to at least r members of $\mathcal{A}$. Then*

$$rH(X_1, \dots, X_n) \le \sum_{A \in \mathcal{A}} H(X_A).$$

The entropy method behind this result goes back to Chung, Graham, Frankl, and Shearer [8]. The theorem can be viewed as a many-set analogue of submodularity: overlapping coordinate sets may be reorganized without losing total entropy.

Exercise 14.4 (Shearer's inequality from the chain rule)**.** Order the coordinates as $1, \dots, n$, expand every entropy by the chain rule, and use the fact that conditioning reduces entropy. Deduce Theorem 14.3. Then compare this proof with repeated uncrossing: what plays the role of a decreasing marginal contribution?

14.3 Discrete convex analysis and M-convexity

Integral points of an integral base polyhedron form an M-convex set. This is not merely terminology: the base-polyhedral inequalities are equivalent to an exchange structure resembling the basis exchange axiom of a matroid. Matroid bases are the 0–1 prototype of this exchange phenomenon; bases of integral polymatroids allow larger nonnegative integer coordinates; general M-convex sets arise from integral base polyhedra even when the underlying submodular function is not monotone. Thus M-convexity continues the exchange idea beyond the polymatroid level.

Theorem 14.5 (M-convex exchange theorem)**.** *Let $B \subseteq \mathbb{Z}^U$ be nonempty. Then B is the set of integer points of an integral base polyhedron if and only if, whenever $x, y \in B$ and $x_u > y_u$ for some u, there exists v with $x_v < y_v$ such that both $x - e_u + e_v$ and $y + e_u - e_v$ belong to B. In this characterization the exchange axiom forces all members of B to have the same coordinate sum; the corresponding base polyhedron is bounded because U is finite.*

This equivalence is a basic theorem of discrete convex analysis; see Murota [47, Chapter 4]. It turns continuous-looking base-polyhedral geometry into a local exchange rule on integer vectors.

A second theorem explains why the quadratic objective that already appeared in our uncrossing potentials is so natural here.

Theorem 14.6 (Frank–Murota decreasing minimization)**.** *Let B be an M-convex set. An element of B is decreasingly minimal—that is, lexicographically minimal after sorting its coordinates in nonincreasing order—if and only if it minimizes the square-sum $\sum_{v \in U} x_v^2$ over B.*

See Frank and Murota [23, 24]. In graph orientation problems, the vectors in B are often indegree vectors, so Theorem 14.6 turns "make the orientation as balanced as possible" into a separable convex optimization problem.

Project 14.7 (Balanced highly connected orientations)**.** Fix a finite loopless graph G that admits a k-arc-strong orientation, and let $\mathcal{D}_k(G)$ be the set of indegree vectors of all such orientations. Determine in small examples whether $\mathcal{D}_k(G)$ is M-convex or is naturally represented as a translate or projection of an integral base polyhedron. When it is, compare decreasingly minimal orientations with the minimizers of the square-sum of the indegrees.

14.4 Coarea, cuts, and threshold rounding

By Proposition 1.17 and Theorem 1.20, graph total variation has the coarea formula discussed earlier. If $x \in [0,1]^V$, then

$$\sum_{uv} w_{uv} |x_u - x_v| = \int_0^1 c(\{v : x_v \ge t\})\, dt.$$

The point of this identity is algorithmic as well as geometric: the right-hand side is the average cost of the threshold cuts.

Corollary 14.8 (Thresholding lemma)**.** *For every $x \in [0,1]^V$, there exists a threshold $t \in [0,1]$ such that*

$$c(\{v : x_v \geq t\}) \leq \sum_{uv} w_{uv}|x_u - x_v|.$$

Proof. If every threshold cut had larger cost than the right-hand side, integrating over an interval of length one would contradict the coarea identity. □

The level-set representation is part of the Lovász-extension viewpoint [41]; the terminology and continuous analogue come from the coarea formula for functions of bounded variation, for which a standard reference is Evans–Gariepy [17].

Exercise 14.9 (When thresholding is exact)**.** Suppose a convex relaxation of a cut problem has graph total variation as its objective and every feasibility constraint is preserved by taking superlevel sets. Prove that thresholding yields an integral solution of no larger objective value. Then give a simple example showing why an additional quadratic term in the objective destroys this immediate averaging argument.

Part V
Projects and further study

15 Research directions and open problems

This section is intentionally exploratory. Statements labelled "open" are dated to August 2026; for fast-moving topics, students should perform a fresh literature check before beginning a project.

15.1 Reconfiguration of highly connected orientations

Ito, Iwamasa, Kakimura, Kamiyama, Kobayashi, Maezawa, Nozaki, Okamoto, and Ozeki proved that the edge-flip graph of k-arc-strong orientations is connected whenever the underlying graph is $(2k+2)$-edge-connected [35].

Research question 15.1 (The $(2k+1)$ threshold)*.* Is the edge-flip graph of k-arc-strong orientations connected for every $(2k+1)$-edge-connected graph?

Ito et al. [35] explicitly left open whether $(2k+2)$ can be weakened to $(2k+1)$ for $k \geq 2$. A natural first project is exhaustive computation for small k and small multigraphs.

Research question 15.2 (Shortest monotone flip sequences)*.* What is the correct worst-case length of a monotone flip sequence that reaches a maximum-connectivity orientation? Can a shortest such sequence be found in polynomial time?

The constructive proof of Ito et al. [35] gives a polynomial bound on the length of the monotone flip sequence; the paper also asks about tightness of that bound and the complexity of finding shortest sequences.

15.2 Infinite Nash–Williams orientations

Nash–Williams conjectured that the finite orientation theorem extends to infinite graphs. Assem, Koloschin, and Pitz proved the optimal $2k$ bound for all locally finite graphs with countably many ends [4].

Research question 15.3 (Infinite orientation problem). Determine the widest natural class of infinite graphs for which every $2k$-edge-connected graph admits a k-arc-strong orientation. In particular, understand the role of uncountably many ends and of non-locally-finite behavior.

This topic is attractive because the finite proof mechanisms split apart in the infinite setting: compactness, ends, topological cycle spaces, and transfinite constructions enter alongside finite uncrossing.

15.3 Can splitting-off be made convex?

Splitting-off proofs are local and combinatorial, while submodular-flow proofs are global and polyhedral.

Research question 15.4. Is there a natural potential function or base-polyhedral path whose elementary moves are precisely admissible splitting-off operations? Can dangerous-set uncrossing be interpreted as a normal-cone calculation for a polyhedron?

A successful answer would conceptually unify the first two proofs of Nash–Williams rather than merely place them side by side.

15.4 Beyond graphs: hypergraphs and set-pair connectivity

Many connectivity requirements in mixed graphs, hypergraphs, and augmentation problems are governed not by fully submodular functions but by *crossing* or *intersecting* supermodular functions.

Project 15.5. Choose one graph theorem in these notes and replace its cut requirement k by a set-dependent demand $p(X)$. Determine the weakest hypothesis on p—fully supermodular, crossing supermodular, intersecting supermodular, or skew-supermodular—under which the uncrossing proof survives.

This leads quickly to Frank's orientation and connectivity-augmentation theorems and to modern survivable network design.

15.5 Algorithmic uncrossing

Uncrossing is often presented existentially. For computation, one wants an efficient method to transform an arbitrary dual optimum into a laminar one.

Research question 15.6. For a given cut-based LP, what is the fastest explicit uncrossing algorithm, and can the laminar certificate be maintained dynamically as the graph changes?

This connects structural graph theory with dynamic algorithms and sensitivity analysis in optimization.

15.6 Submodular minimization inside graph algorithms

General submodular function minimization is polynomial-time solvable, but graph cut functions admit much faster specialized algorithms.

Project 15.7. Take a theorem whose proof uses a generic "choose a minimal tight set" step. Compare three implementations:

1. generic submodular function minimization;
2. a max-flow/min-cut oracle;
3. a specialized graph algorithm exploiting symmetry or laminarity.

Quantify the gap in running time and identify the structural reason for it.

16 A Menger follow-up project menu: paths, cuts, metrics, and beyond

The Menger material offers a natural collection of after-class projects because it looks at the same theorem from a different center of gravity than the main body of these notes. In the present handout, Menger is one stage in a larger story about submodularity, tight sets, uncrossing, convexity, branchings, tree packings, and orientations. The Menger-focused study-report slides from 2025, available at `https://math.sjtu.edu.cn/faculty/ykwu/data/Talk/20250829.pdf`, invite the complementary question: what further mathematics radiates outward from Menger's theorem itself?

The projects below are deliberately modular. An individual student or a group should normally choose *one* track and develop it substantially. Different groups may choose different tracks. There is no expectation that every participant read the same research papers or cover the whole menu.

Common entry task

Before specializing, read the Menger slides together with Sections 3 and 4. Prepare a short comparison map with three columns:

Theme	**Role in these notes**	**Role in the Menger material**
packing–covering	one instance in a wider uncrossing theory	the organizing min–max principle
residual augmentation	proof of Menger/max-flow–min-cut	a model algorithmic proof technique
submodularity	organizes tight cuts and later graph theorems	explains the geometry of minimum cuts
convexity/duality	Lovász extension, coarea, zonotopes	path-packing LPs, metric duals, threshold cuts
what comes next	Edmonds, Frank, Tutte, orientations	matching, ears, gammoids, multiflow, topology, infinity

The comparison need not use exactly these words, but it should identify at least one point where the two sources genuinely complement one another rather than merely repeat the same theorem.

Track A: vertex Menger, transformations, and endpoint conventions

Develop the vertex and edge forms of Menger's theorem for two disjoint terminal sets. Give a complete proof of at least one reduction between the two theorem schemes, for example vertex splitting or the augmented line-graph construction. Include examples showing that vertex and edge connectivity can differ greatly in the same graph, and explain carefully the conventions needed when two local terminals are adjacent.

A strong report should answer: *which aspects of Menger are invariant under the reductions, and which geometric or graph-theoretic properties are not preserved?* Standard references include Diestel [11] and Schrijver [60].

Track B: Hall, Kőnig, fans, ears, and structural consequences

Start from the exercise in Section 4 deriving Hall's deficiency formula from set-to-set Menger. Then choose two or three further consequences of vertex Menger—for example Kőnig's theorem, the fan lemma, Whitney's connectivity characterization, or open ear decompositions—and prove them completely.

The goal is not to collect implications mechanically. Organize them around the question: *how does a path–separator min–max theorem force global graph structure?* For an ear-decomposition project, identify precisely where two disjoint attachment paths are needed and why an ordinary connectivity argument would not suffice.

Track C: Menger and gammoids

Study the matroid obtained by declaring a set of vertices independent when its members can be linked by pairwise vertex-disjoint directed paths to a prescribed set of sinks. Explain how Menger's theorem supplies the exchange mechanism behind this construction and how the resulting *gammoid* fits beside the transversal-matroid viewpoint arising from Hall's theorem.

A suitable report should define strict gammoids carefully, prove at least one nontrivial matroid property from a linkage argument, and compare the relevant rank/separator formula with the submodular rank functions discussed earlier in these notes. For background see Oxley [51] and the treatments of linkage matroids in Schrijver [60].

Track D: from one commodity to multiflow and metric duality

Begin with the four-cycle with opposite demand pairs and verify explicitly that the natural cut condition can hold although the two prescribed pairs cannot be routed by edge-disjoint paths. Formulate a fractional multicommodity-flow LP and derive its dual. Explain why the dual variables naturally produce an edge-length function and a shortest-path metric rather than a single cut.

Then return to the one-commodity case and show how thresholding a shortest-path potential collapses the metric certificate back to a cut. Compare this argument with the coarea/thresholding proof in Section 2.5. The central question is:

> Why is one commodity special enough that a cut is a complete dual certificate, while several prescribed commodities naturally require a larger metric dual space?

Frank [22] and Schrijver [60] provide broad references.

Track E: planar multiflow and the Okamura–Seymour phenomenon

Read Okamura and Seymour [50]. State their planar multiflow theorem precisely and explain why it is striking in light of Track D: under a strong planar terminal-position hypothesis, cut conditions recover much of the exactness that fails for general multiflow.

It is not necessary to reproduce the entire original proof. A good project may instead prove a meaningful special case, reconstruct one central lemma, or analyze carefully why the common-face hypothesis matters. Include examples showing what fails when the hypothesis is removed.

Track F: Gomory–Hu trees—all pairs, but not all at once

Read Gomory and Hu [31]. Explain why a weighted tree on the original vertex set can encode all pairwise minimum-cut values of an undirected capacitated graph, while not solving a simultaneous multiflow problem.

The mathematical core of the project should go beyond stating the theorem. Prove at least one structural lemma in the construction, preferably one showing how minimum cuts can be selected compatibly or uncrossed. Work out a nontrivial example by hand; a computational implementation and visualization of the fundamental cuts is an optional extension. Relate the resulting nested cut structure to the minimum-cut lattice and laminarity developed earlier in the handout.

Track G: planar duality, grids, and topology

Investigate how an edge cut in a plane graph appears as a cycle or closed curve in the planar dual. Prove a precise planar cut–cycle statement, then apply it to a rectangular grid or to a Hex-type crossing alternative. Explain why vertex separators require more care than edge cuts.

A stronger project can examine what breaks on a surface of positive genus: a nonseparating dual cycle need not correspond to a primal cut. This provides a concrete way to see topology entering a theorem that, in the plane, looks purely combinatorial.

Track H: infinite Menger and orthogonal certificates

The finite proofs in this handout use finiteness essentially. Read Aharoni and Berger [3] and compare their infinite theorem with the finite statement. The emphasis should be on the strength of the certificate: the infinite theorem constructs a disjoint path family and a separator choosing exactly one vertex from each path, not merely an equality of two cardinal numbers.

A good report should identify several precise places where the finite arguments fail—for example termination of augmentation, compactness of a finite path LP, or naive maximality—and explain why the orthogonality conclusion is the correct infinite replacement.

Possible deliverables and expectations

The scope should be matched to the chosen track rather than fixed uniformly. A typical project may consist of a self-contained written report, a short presentation, or both. Whatever the format, the mathematical contribution should normally contain:

1. at least one theorem or substantial lemma proved completely by the students;
2. at least one worked example, counterexample, or explicit construction;
3. a clear comparison with the Hall/Menger/submodularity material in this handout;
4. a small annotated bibliography explaining what was learned from the principal sources;
5. an explicit distinction between results proved in the report, results quoted from the literature, and questions or extensions proposed by the students.

For a group project, the final report should read as one coherent mathematical exposition rather than a concatenation of independently written summaries.

A unifying question for all tracks. Menger's theorem is unusually exact because the packed objects, the obstruction, the algorithm, and the convex dual all fit together perfectly. Each project above asks what happens when one ingredient is changed. A useful diagnostic is always:

> What is being packed? What is the obstruction? What is the correct dual object? Where do integrality, submodularity, or topology enter? If exact min–max equality fails, which part of the Menger architecture is responsible?

17 A companion project to Hall: stable marriage, Galvin, and the Dinitz problem

The 2022 slide deck, *Dinitz conjecture and the Shaw prize 2022*, available at `https://math.sjtu.edu.cn/faculty/ykwu/data/Talk/20220925.pdf`, makes an unusually natural companion to Section 3. Hall's marriage theorem asks whether a bipartite graph contains a matching saturating one side. The stable-marriage problem keeps the same two-sided matching language but adds *preferences*: a matching is now required not merely to exist, but to have no pair of participants

who would both rather leave their assigned partners and match with one another. The slides then use this stability mechanism in a strikingly different problem—Galvin's proof of the Dinitz conjecture on list edge coloring—before continuing to choosability and related questions.

This gives a coherent second after-class project with a different flavor from the Menger menu. The Menger project develops packing versus separation. The present project develops the progression

$$\begin{aligned}\text{Hall} &\longrightarrow \text{Kőnig edge coloring} \longrightarrow \text{Gale–Shapley stability}\\ &\longrightarrow \text{Galvin's list-edge-coloring theorem.}\end{aligned}$$

The most important point is not the shared word "marriage". Hall and Gale–Shapley solve different feasibility problems, and the project should make that distinction precise.

A short classroom bridge after Hall

A useful introduction takes only one small preference table. Let $M = \{m_1, \dots, m_n\}$ and $W = \{w_1, \dots, w_n\}$. Each person has a strict preference order on the opposite side. For a perfect matching μ, a pair $m_i w_j \notin \mu$ is a *blocking pair* if m_i prefers w_j to $\mu(m_i)$ and w_j prefers m_i to $\mu(w_j)$. The matching is *stable* if it has no blocking pair.

Run the proposer version of deferred acceptance on a three-by-three example: every currently unmatched proposer applies to the most preferred receiver who has not previously rejected that proposer; every receiver keeps the most preferred proposal seen so far and rejects the others. Students should record the proposal history rather than only the final matching. The history is the certificate used in the proof of stability.

Theorem 17.1 (Gale–Shapley, 1962)**.** *For two finite sets of the same size with strict complete preference orders, the deferred-acceptance algorithm terminates and produces a stable perfect matching.*

The common first assignment is to prove this theorem from the algorithm. Termination follows because no proposer makes the same proposal twice. For stability, if m prefers w to his final partner, then m must have proposed to w earlier and been rejected; from that moment onward w holds partners she weakly improves, so she cannot prefer m to her final partner. Students should also explain why, with equal sides and complete preference lists, nobody is left unmatched. Gale and Shapley's original paper is still an exceptionally readable source [29].

Hall versus stable marriage: same actors, different obstruction

It is worth making the comparison explicit. In Hall's theorem the data are acceptable edges and the goal is a matching saturating one side. The obstruction is a deficient set X with $|N(X)| < |X|$. In stable marriage the new data are preference orders, and the local obstruction to a proposed matching is a blocking pair. In the complete balanced model the underlying graph is $K_{n,n}$, so Hall's condition is automatic; the content is entirely in stability and the dynamics of rejection.

The distinction becomes sharper with incomplete preference lists. Treat being unmatched as worse than every acceptable partner and run deferred acceptance only along acceptable edges. A stable matching still exists, but a perfect matching in the acceptability graph need not imply the existence of a *perfect stable* matching. As a short exercise, students should find the smallest example in which the acceptable bipartite graph satisfies Hall's condition but every stable matching leaves somebody unmatched. This prevents a common but incorrect impression that stable marriage is simply Hall's theorem with rankings added.

A second comparison question is algorithmic. Hall's theorem can be proved by augmenting paths or by the tight-set induction used earlier in these notes. Deferred acceptance has a different

monotonicity: proposals move down one side's preference lists, while the partners tentatively held by the other side only improve. Ask students to identify exactly where each monotonicity enters its proof.

Track A: the structure of stable matchings

After proving existence and stability, continue to the classical extremal property of deferred acceptance: when M proposes, every proposer receives the best partner that proposer can obtain in any stable matching, while the receiving side simultaneously receives its worst stable partner. A good project should prove the proposer-optimal statement carefully from a "first rejection that could occur in a stable matching" argument.

A more ambitious continuation is the lattice of stable matchings. The stable matchings of a fixed instance admit a natural distributive-lattice structure when ordered by the preferences of one side; rotations give an efficient way to describe all stable matchings. Gusfield and Irving [32] are a standard reference. This makes a useful comparison with the minimum-cut lattice from Section 4: both are lattices of optimal or feasible combinatorial objects, but the stable-matching lattice operations are induced by preference order and should not be confused with ordinary union and intersection of edge sets.

Suitable after-class questions include determining when the stable matching is unique, constructing instances with several stable matchings, and comparing the outcomes when the proposing and receiving sides are interchanged. Small computational experiments are useful here, provided the report explains the theorem suggested by the data rather than merely enumerating matchings.

Track B: from Hall to Kőnig to Galvin—the mathematical core of the slides

The "training camp" problem in the slides can be translated directly into list edge coloring. Boys and girls form the two vertex classes of a complete bipartite graph; a mixed pair is an edge; the coaches willing to take that pair form the edge's list of available colors. Assigning every pair to one coach so that no person appears twice in the same camp is exactly a proper edge coloring chosen from the prescribed edge lists.

For $K_{n,n}$, the Dinitz conjecture says that lists of size n always suffice. Galvin proved the stronger theorem that every bipartite multigraph G satisfies $\chi'_\ell(G) = \chi'(G) = \Delta(G)$ [30]. The stable-marriage theorem enters his proof in a very concrete way.

The project can reconstruct the proof in four steps.

1. First derive Kőnig's edge-coloring theorem for bipartite graphs from Hall's theorem. One convenient route is to embed a bipartite graph of maximum degree Δ into a Δ-regular bipartite multigraph, use Hall to find a perfect matching, delete it, and iterate. Thus Hall supplies a proper Δ-edge-coloring ϕ.

2. Let $G = (X, Y; E)$ and orient its line graph using ϕ. If two incident edges e, e' satisfy $\phi(e) > \phi(e')$, orient from e to e' when they meet in X, and orient from e' to e when they meet in Y. For an edge with $\phi(e) = r$, the outdegree is at most $(r-1) + (\Delta - r) = \Delta - 1$.

3. Prove that every induced subdigraph of this orientation has a kernel. Translate an induced subgraph of the line graph back into a bipartite subgraph of G. Vertices of X prefer incident edges with smaller ϕ-value, while vertices of Y prefer larger ϕ-value—exactly the opposite preference orders dramatized by the Latin-square table in the slides. A Gale–Shapley stable matching is then an independent set in the line graph, and stability says that every vertex outside it has an outgoing arc into it. Hence the stable matching is a kernel.

4. Prove the kernel/list-coloring lemma: if a digraph is kernel-perfect and every vertex v has a color list of size at least $d^+(v) + 1$, then the underlying graph is list-colorable. Induct by choosing a color c, coloring a kernel among the vertices whose lists contain c, deleting that kernel, and deleting c from the remaining lists. The kernel property guarantees that whenever a list loses one color, the relevant outdegree also falls by at least one.

Putting the four steps together gives Galvin's theorem and therefore Dinitz. This route also explains the Latin-square preferences in the slides rather than treating them as a clever trick: a Latin square is simply a proper n-edge-coloring of $K_{n,n}$, and the "smaller on one side, larger on the other" rule is precisely the preference system that makes the Galvin orientation kernel-perfect.

Students should also work through one of the explicit four-by-four training-camp examples from the slides by hand. The purpose is to see the proof mechanism before compressing it into the line-graph language. Janssen's earlier rectangle result [37] and Zeilberger's exposition of Galvin's proof [65] are useful historical companions.

Track C: follow the slides from Dinitz to choosability

The last part of the slides changes viewpoint from edge lists on a bipartite graph to list coloring of ordinary graphs. Students should first verify on the displayed $K_{2,4}$ example that the chromatic number can be smaller than the choice number. Then translate Dinitz into the line-graph statement that $L(K_{n,n})$ is n-choosable, and explain why Galvin's theorem proves the corresponding statement for the line graph of every bipartite multigraph. The slides also place this inside the historical List Coloring Conjecture for line graphs, which appeared in print in work of Bollobás and Harris [5]. The early framework for choosability is due to Vizing and to Erdős, Rubin, and Taylor; see, for example, Erdős–Rubin–Taylor [16].

For a computational or exploratory continuation, the cube and generalized Hamming graphs in the final slides make good test cases. Determine the choice number in the smallest instances that can be handled exactly, construct list assignments witnessing lower bounds, and compare ordinary coloring with list coloring. A good report should distinguish a proved theorem from a pattern seen only in computation.

An optional historical branch is to compare the combinatorial stable-marriage proof with the algebraic route surrounding the Combinatorial Nullstellensatz. Janssen obtained the rectangle case before Galvin's theorem, while Galvin's 1995 argument settled the bipartite list-edge-coloring problem in full [37, 30]. The point of such a comparison is to ask what each proof method makes visible, not merely to list two proofs of the same theorem.

Track D: a convex-geometric companion—the stable-marriage polytope

Encode a perfect matching by its incidence vector in edge space and take the convex hull of all stable matchings. Vande Vate gave a linear description whose extreme points are stable matchings [63], and Rothblum gave a streamlined and more general polyhedral treatment [56].

A suitable project should write down the matching equations and the stability inequalities for a small instance, compute the resulting polytope, and verify integrality directly in that example. Then compare this polytope with the matroid base polytopes and submodular base polyhedra from earlier chapters. The comparison should be precise: the stable-marriage polytope is an integral combinatorial polytope, but it is not being introduced here as a polymatroid base polytope. The interesting common theme is instead that a combinatorial condition—here "no blocking pair"—has a faithful system of linear inequalities and an integral convex hull.

Suggested after-class organization

The project works well if everybody first completes a small common core and then chooses one direction. The common core should contain: one hand execution of deferred acceptance; a complete proof of the Gale–Shapley theorem; one explicit comparison with Hall's theorem, including the incomplete-list counterexample; and a translation of the training-camp problem into list edge coloring. After that, a student or group can concentrate on the lattice structure of stable matchings, Galvin's proof, choosability/Hamming-graph experiments, or the stable-marriage polytope.

A finished report should normally contain at least one theorem proved completely, one worked preference or list-coloring example, and one explicit bridge back to the lecture notes. For the Galvin track, the preferred bridge is the chain "Hall → Kőnig → Gale–Shapley → kernel → list coloring". For the lattice track, compare the stable-matching lattice with the minimum-cut lattice. For the polyhedral track, compare the stability inequalities with the supporting-inequality viewpoint developed for base polyhedra.

A particularly effective presentation begins with a two- or three-by-three preference table rather than with the final theorem. After the audience has watched deferred acceptance create a stable matching, the presenter can ask the key question from Galvin's proof: *why should a stable matching of edges become a kernel in an oriented line graph?* Answering that question turns the matchmaking story into graph theory.

What this project should teach. Hall's theorem says when a static family of acceptable pairs contains enough disjoint edges. Gale–Shapley shows that ordered preferences support a different kind of existence theorem, driven by monotone rejection rather than a subset-counting obstruction. Galvin's proof is remarkable because it transports that stability theorem into list coloring. A method created for matching people becomes a kernel certificate in a line graph, and that certificate solves Dinitz's coloring problem.

18 Selected exercises with hints

Exercise 18.1 (Submodular slack)**.** For an undirected graph define $\sigma(X,Y) := d(X) + d(Y) - d(X \cap Y) - d(X \cup Y)$. Prove that $\sigma(X,Y)$ equals twice the number of edges between $X \setminus Y$ and $Y \setminus X$.

Hint. Count each edge according to which of the four Venn regions contain its endpoints.

Exercise 18.2 (Directed slack)**.** Find an analogous explicit formula for $\rho(X) + \rho(Y) - \rho(X \cap Y) - \rho(X \cup Y)$ in a digraph.

Exercise 18.3 (Canonical minimum cut)**.** Fix s, t. Prove that the intersection of all minimum t-sides is the unique inclusion-minimal minimum t-side. Give a max-flow interpretation of this set in terms of the residual graph of a maximum flow.

Exercise 18.4 (Tutte from rooted orientations)**.** Assume the theorem: an undirected graph has an orientation with $\rho(X) \geq k$ for every nonempty $X \subseteq V \setminus \{r\}$ if and only if $e(\mathcal{P}) \geq k(|\mathcal{P}| - 1)$ for every partition $\mathcal{P}$. Deduce Theorem 7.1 from Edmonds' theorem.

Exercise 18.5 (Fractional Nash–Williams)**.** In the convex proof of Theorem 8.2, verify carefully that $x \equiv 1/2$ satisfies every cut inequality. Explain why this argument would fail if the undirected connectivity assumption were only $(2k-1)$.

Exercise 18.6 (A polyhedral proof of convexity)**.** Using Theorem 1.14, prove directly that the Lovász extension is the support function of $B(f)$. Then compute $B(f)$ explicitly when $f(X) = \min\{|X|, r\}$.

Exercise 18.7 (Cut base polyhedron)**.** For a weighted undirected graph, characterize vectors in the base polyhedron of the cut function. Explore their interpretation as divergences of edge vectors bounded by capacities.

Exercise 18.8 (Experimental edge flips)**.** For $k = 2$, enumerate all orientations of a small 4-edge-connected multigraph. Build the graph whose vertices are 2-arc-strong orientations and whose edges are single edge flips that preserve 2-arc-strong connectivity. Is it connected? Repeat for graphs of connectivity 5 and 6.

19 Closing perspective

The eight orange uncrossing case studies and six violet bridge points provide two summaries of the course. The first summary follows how tight obstructions are simplified; the second follows how the same combinatorial objects acquire convex-geometric incarnations. A useful final exercise is to revisit each orange box and ask which, if any, violet bridge explains the same phenomenon polyhedrally.

The central lesson is not that submodular functions provide one more technique for graph theory. Rather, they explain why a large class of graph-theoretic proofs have the same shape, and why apparently different continuous constructions—Choquet integration, support functions, normal fans, and Euclidean projection—keep returning the same discrete sets.

A connectivity theorem begins with exponentially many inequalities indexed by vertex sets. Minimal obstructions are tight. Tight objects cross. Submodularity allows them to be uncrossed. Uncrossing produces laminar structure. Laminar structure supports induction, decomposition, integrality, or a safe local move. When this pattern is translated into convex language, the same theorem appears as a statement about base polyhedra, support functions, normal cones, and integral points. When translated into integration language, a vector is decomposed into nested level sets and the same set function is integrated as a Choquet integral. When translated into Euclidean geometry, the nearest point of a base polyhedron can reveal a minimizer by its sign pattern.

This is why Hall's theorem, minimum cuts, arborescences, spanning-tree packings, splitting-off, graph orientations, and submodular flows belong naturally in one lecture course. Their surface statements are different; their geometry is remarkably similar. The hierarchy developed in Part I provides one final way to organize them: matroid exchange sits inside polymatroid diminishing returns, while graph cuts and submodular flows show how the same uncrossing geometry persists after monotonicity has disappeared.

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